SECTION - 18

TECHNICAL SPECIFICATION OF WORKS

PART- II (A)

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TECHNICAL

PART- II (A)

1.0 SCOPE

This Specification provides for collection of the materials supplied by the employer from employer's stores, procurement and supply of balance materials to complete

the work, transport of all materials to site, execution of civil works in the switchyard, construction of control house, erection of structures, bus-bars, equipment in switchyard and control room etc. setting to work and for performance guarantee of EHV sub-stations and/or bay extensions. The contractor shall execute the works to the extent required to complete the work as a whole for substation/Bay.

The works include:

1.1 WORKS IN THE SWITCHYARD

i) Excavation for main and equipment structure foundations, plinths for Transformers, gate pillars, levelling, cable laying, earthing etc.

ii) RCC/CC foundations for structures, Transformer and circuit breaker plinths etc.

iii) Back filling for foundations

iv) Levelling of sub-station site with borrowed gravel or excavated earth including clearing of site.

v) Construction of compound wall and retaining wall

vi) Construction of cable duct including cable ducts crossing Transformer track

vii) Formation of roads

viii) Spreading of HBG metal

1.2 CONSTRUCTION OF CONTROL HOUSE

The work involves excavation and laying of foundations, construction of super structure, laying RCC slab, providing doors, windows, flooring with ceramic tiles, painting, providing water supply and sanitary arrangements etc., including electrification of control house.

1.3 ERECTION WORKS

i) Erection of main and auxiliary structures including supply of raw steel, fabrication and galvanization of structures and equipment including panels.

ii) Erection of all substation/bay equipment including Panels, Capacitor Banks etc., but excluding Power Transformers.

iii) Stringing of bus bars, jumpering, connections between equipment, buses, etc.

iv) Stringing of ground wire

v) Earthing system including laying of C.I. pipes and B.H. coke for earth pits

vi) Cable laying and terminations including erection of marshalling kiosk/boxes.

vii) Yard illumination, fire fighting etc.

1.4 ELECTRIFICATION OF CONTROL HOUSE (Two storeyed Building):

The work involves electrification of control house including supply of all the materials, wiring etc.

2.0 TYPE OF CONSTRUCTION

The sub-station structures shall be lattice type galvanized structures. The foundations of 220KV and 132KV main structures are stub type and other structures are foundation bolt type. The 220KV bus is strung with MOOSE ACSR conductor and 132KV, 33KV buses are strung with ZEBRA ACSR conductor. A control house to locate panels, battery, battery charger and D.C distribution board is to be constructed as indicated in Clause 1.2 above.

2.1 DRAWINGS

The following drawings are enclosed to give an idea about the work involved :

i) Control house plan, foundation details, Reinforcement details of columns, beams, roof, Details of Doors and windows, Electrification of control house, cable duct in control house etc.
ii) Compound wall
iii) Cable duct in switchyard, cable duct at road crossing points.
iv) Earth mat details
v) Main gate and wicket gate
vi) Foundation and structural drawings
vii) Retaining wall

3.0 CODES AND STANDARDS

All equipment materials fabrication galvanising and tests shall conform to the latest applicable standards, codes etc. mentioned here in or to equivalent applicable international standards approved by the engineer

Sl. No.	STANDARD	TITLE
1.	IS 8112/1989	43 grade ordinary portland cement
2.	IS 1786/1985	High strength deformed steel bars and wires for concrete reinforcement
3.	IS 456/1978	Code of practice for plain and reinforced concrete
4.	IS 383/1970	Course and fine aggregates from natural sources for concrete

5.	IS 2062/1992	Steel for general structural purposes
6.	IS 432(Part-1& / Part-2)/1982	Mild steel and medium tensile steel bars and hard drawn steel wire for concrete reinforcement.
7.	IS 12427/1988	Transmission Tower Bolts
8.	IS 1367 (part II)	Technical supply conditions for threaded steel fasteners - Product grade and tolerances
9.	IS 1363 (part III)	Hexagon Nuts(size range M5 to M64)
10.	IS3063/1994	Fastners - single coil rectangular section spring washers
11.	IS2629/1985	Recommended practice for Hot dip galvanizing on Iron & Steel
12.	IS4759/1984	Hot dip Zinc coatings on structural steel and allied products
13.	IS7181/1986	Horizontally C.I double flanged pipes for water, gas and sewage valve fittings for compressed gas
14.	IS3224/1979	Cylinders excluding liquified petroleum gas cylinders
15.	IS2026	Station Transformer
16.	IS5039/1983	Distribution pillars for voltages not exceeding 1000 volts
17.	IS4072/1975	Steel for spring washers
18.	IS1573/1986	Electroplated Coatings of Zinc on Iron & Steel
19.	IS2202(Part-I)/1991	Wooden flush door shutters
20.	IS4351/1976	Specification for steel Door frames
21.	IS513/1994	Coldrolled low carbon steel sheets and strips
22.	IS617/1975	Alluminium and alluminium alloy ingots and castings for general engineering purposes
23.	IS5561/1970	Electric power connectors

24.	IS486(Part-I)/ 1993	Metal fittings of Insulators for overhead power lines with normal voltage greater than 1000V - General requirements & Tests
25.	IS2486(Part-2)/ 1989	Insulator fittings for overhead power lines with a nominal voltage greater than 1000V - Dimensional requirements
26.	IS2108/1977	Blackheart malleable iron castings.
27.	IS2486/	Hardware fittings, clamps and connectors
28.	IS4984	P.V.C. Pipes
29.	APSS	Andhra Pradesh Standard Specification.

List of Approved Brands / makes for procurement of materials:

Sl. No.	Description of Material		Approved brands
1	Cement	:	**1**) ACC Limited **2**) Kesoram Cements **3**) Orient Cements **4**) Zuari Cements **5**) CCI Limited **6**) Andhra Cements **7**) Coramandal Cement **8**) Raasi Cement **9**) Sri Vishnu Cements **10**) Madras Cements **11**) Ultra Tech Cement Limited – APCW **12**) KCP Limited **13**) Penna Cements **14**) Panyam Cement **15**) Grasim **16**) Rajashree Cement **17**) Mysore Cement **18**) Century Cement **19**) Ambhuja Cement , **20**) Priya Cement, **21**) Duncan cement, **22**) Parashakthi cement, **23)** M/s. Nagarjuna Cements, **24)** M/s. JSW Cements, **25)** M/s. Bharathi Cements, **26)** M/s. Sagar Cements, 27) Deccan Cement and **27**) Maha cement.
2	Reinforcement and structural steel	:	**1**) VSP **2**) SAIL and **3**) TISCO
3	Paints		
	a) External Walls	:	**1**) Asian **2**) Berger **3**) Nerolac **4)** Nexon **5)** Neon
	b) Internal Walls & Oil paints	:	**1**) Asian **2**) Berger **3**) ICI **4**) Nerolac **5**) Dulux **6**) Jenson & Nicholson
4	Aluminium Profiles	:	Jindal and Hindalco
5	Water supply and sanitary works	:	**1**) Hindware, Parry, Neycer for washbasins, water closets and urinals. **2**) Tata / Jindal / Zenith for GI pipes **3**) Jaguar stopcocks for control room, Richo & Seiko taps for earth pits **4**) Sudhakar & Prince for PVC pipes 5) Sintex water tank
6	Bore Well	:	**1**) Texmo **2**) Crompton **3**) Suguna **4**) CRI and **5**) Lakshmi for motor
7	Electrical works	:	**1**) Fine cab, poly cab, V-guard cables for wiring and switch board connections. **2**) Orient, Khaitan, Usha, Havells make ceiling and exhaust fans. **3**) Osram, Surya for tube lights & incandescent bulbs **4**) Philips / Crompton greaves for decorative portico lights & sodium vapor lamps **5**) Crompton Greaves, Osram, Sury batten type holder and 100 watts bulb. **6**) Achor 15 / 5APower socket 7) 5A 2/3 socket with steel box **8**) 4 Way distribution box of sputhnik,Standard and with MCB's. **9**) 32 amps ICTP switch of MDS / INDOCAP

4.0 CLIMATIC & ISOCERAUNIC CONDITIONS

The climatic and isoceraunic conditions where the sub-station work/bay extensions are carried out are as given below

(i)	Annual rain fall (maximum)	100 cm
(ii)	Maximum ambient temperature	
	(a) In shade	40 deg. C
	(b) In sun	45 deg. C
(iii)	Minimum temperature	10 deg. C
(iv)	Maximum relative humidity	74 %
(v)	Minimum relative humidity	10 %
(vi)	Number of Monsoon months	4
(vii)	Average Number of rainy days per annum	90
(viii)	Average annual rain fall	745 mm
(ix)	Wind speed	44m/sec to 50m/sec
(x)	Average Number of thunder storm days per annum	50
(xi)	Maximum altitude above sea level	1000 m

5.0 COMPLIANCE WITH REGULATIONS

Unless otherwise specified works shall be carried out in accordance with the Indian Electricity Act. 1910, Indian Electricity Rules 1956 or any other revisions thereof which may be issued during the currency of the contract and the requirements of any other regulations and Acts in India to which the employer may be subjected to.

5.1 The contractor himself has to arrange for the permits required for the operation of the vehicles used in construction works and such other permits not being arranged by the employer/Engineer.

6.0 MATERIALS TO BE SUPPLIED BY THE APTRANSCO AND TO BE PROCURED BY THE CONTRACTOR

The materials supplied by the Employer and procurement of materials by the contractor are mentioned in Appendix-II for the erection of sub-station/bay extension separately. The complete technical and other particulars will be intimated to the successful bidder after his bid is accepted. The contractor shall be fully responsible for the proper handling, maintenance and safe custody of the materials given to him or his authorised representative at the employer's stores for erection from the time of taking over till the complete sub-station/bay extension is taken over by the employer.

6.1 SURPLUS MATERIALS

On completion of works all materials left surplus from those supplied by the employer under the contract for erection shall be handed over to the employer at the employer's stores in good condition. The contractor shall furnish a statement giving full details of all materials received from the employer, quantities utilized on works (based on the drawings given to him) and the balance returned to the employer and obtain a clearance certificate from the Executive Engineer, with in one month after completion and handing over of the substation/bay extension. It may be noted that returning of all materials and rendering of accounts for the materials received is a pre-condition for release of payment against final bill and also the deposit towards performance at a later date.

6.2 REPLACEMENT

In the event of the material handed over to the contractor for the execution of this contract being lost, damaged or destroyed while being in the custody of the contractor before being taken over by the employer, the contractor shall be liable to make good the loss. In case the contractor fails to return the balance surplus materials, the cost of materials will be recovered from the contractor's bills in accordance with the provisions in employer's Commercial Accounts Manual.

7.0 MATERIALS REQUIRED FOR THE WORK

The requisite quantities of materials specified in Appendix-II required for the completion of sub-station/bay extensions will be handed over to contractor at the Employer's stores indicated, progressively based on the program of the contractor and the actual progress obtained. If additional quantities of materials (in excess of those to be issued as per contract) become necessary due to damage or loss on account of mishandling and negligence of the contractor to complete the work, the contractor shall requisition such quantities duly explaining the circumstances under which the additional quantity has become necessary. Such materials shall be issued by the Engineer only on the written requisition from the contractor and the costs of such materials calculated in accordance with the provisions in employers Commercial Accounts Manual, will be recovered from the payments to be made to the contractor. The contractor will be responsible for the proper handling and maintenance of the materials from the time of receipt by him up to the time of handing over of the completed works to the employer and return of surplus material if any, at the employer's stores. The cost of such handling and storage shall be covered in the quoted prices detailed in bill of quantities. On completion of the works all surplus materials shall be handed over to the engineer at the employer's stores. This shall be done with in a maximum period of one

month after completion and handing over of the sub-station/bay extension works. If the materials are not returned with in this period and/or in the opinion of the engineer are not in a fit condition for use, such surplus materials will be treated as sold to the contractor at the rate calculated in accordance with the provisions in employer's commercial accounts manual and recoveries made accordingly. The retention amount against the contract will be released only after all the balance materials are handed over at employer's stores in proper shape and material accounts are rendered.

8.0 SUPPLEMENTAL ITEMS:

8.1 The quantities indicated in the schedule of the prices are only provisional and are likely to change during actual execution. When quantities of work of any item are likely to exceed the scheduled quantity, the Contractor shall bring the fact to the notice of CE/Projects three weeks in advance and take orders for going ahead with the work. Without approval of this office, the Contractor shall not go ahead with the work wherever there is increase in quantities.

8.2 The contractor is bound to execute all supplemental items that are found essential, incidental and inevitable during execution of main works, at the rates to be worked out as detailed below:

(a) Supplemental items directly deducible from similar items in the original agreement:
The rates shall be derived by adding to or subtracting from the agreement rates of such similar items, the cost of the difference in quantity of material or labour between new item and the similar item in the agreement worked out with reference to the schedule of rates adopted in the sanctioned estimate with which the tenders were compared plus or minus overall tender percentage.

(b) New Items :

(i) Similar items the rates for which cannot be directly deduced from the original agreement.

(ii) Purely new item which do not correspond to / any item in the agreement. The rate shall be estimate rate plus or minus overall tender percentage.

NOTE : The term estimate rate used in (i) and (ii) above means the rate corresponding to the respective item in the sanctioned estimate whose value is shown in the Schedule – A while calling for the tenders for the work or if no such rate is available in the estimate, the rate derived with reference to the schedule of rates adopted in the sanctioned estimate.

(c) Addition of provision towards importation of labour, labour amenities, dewatering etc., in working out supplemental items.

If the new item is in substitution of an old item which allowed for importation of labour, labour amenities, dewatering etc. Those factors may be taken into account in computing the substituted items also at the same rates at which they were originally provided.

In respect of the new items the case has to be considered on its merits and provision for importation of labour, labour amenities, dewatering etc., has to be fully justified.

(d) Execution of items of work in excess of quantities in Schedule 'A' , bill of quantities of tender:

The rates quoted by the tenderer shall hold good irrespective of quantities given in bill of quantities. However, for quantities exceeding 25% over those given in Bill of Quantities, approval of competent authority shall be obtained before executing such quantities against those items**.**

9.0 CLEARING-UP THE SITE:

During construction the contractor shall keep the work site and storage area used by him free from accumulation of waste materials or rubbish and before completing the works the contractor shall, at his own cost, remove or dispose off in a manner satisfactory to the engineer in-charge. All temporary structures, waste and debris shall be cleaned, all holes in the ground shall be filled in and the land restored to its original state as far as practicable and the entire premises should be in a neat and tidy condition of cleanliness as the Engineer may direct. Any damage done to the permanent or temporary work of the Department by the contractor or his sub-contractor shall be made good at contractor's expenses.

10.0 SITE STORES:

The successful bidder shall establish a temporary store at his cost at the sub-station site for storing cement, equipment such as panels etc. This store should be dismantled and site cleared after the work is completed by him.

11.0 SITE CONDITIONS:

The bidder is expected to familiarize with the site conditions and facilities available before quoting. Regarding water supply and electric power supply arrangements, Clause 16 of part-II of technical specification is applicable.

12.0 LEVELLING:

Substation switchyard should be leveled as per directions of Engineer with the surplus earth left over after back filling structure foundations. If surplus earth is not adequate for levelling, the levelling should be done by excavating earth in any portion of substation site whatever type of soil is met with and filling should be done with such earth for levelling switchyard. If this is not sufficient, the switchyard is to be leveled by supplying gravel from the approved borrowed areas. Clearing of the entire site by removing bushes, hedges, unwanted plant growth and other things are in the scope of levelling work. The work shall be carried out as per IS 301,302 and 138.

COMPACTION: Compaction shall be achieved by the use of smooth wheeled rollers, pneumatic type rollers, sheep's foot rollers, vibrating plates, frog rammers, power rammers and such other equipment as shall be specified. Compaction shall be carried out on each layer of thickness not more than 600 mm. Density tests shall be made on each compacted layer and the density attained shall not be less

than 95% of maximum dry density (standard proctor for the type of material used). A record of the density tests conducted on each layer shall be recorded in a register maintained by field staff.

12.1 **ANTI WEED TREATMENT AND ANTI TERMITE TREATMENT**:-
Anti weed treatment and anti termite treatment for excavation in the site shall be done as per Technical Specification.

13.0 EXCAVATION & FOUNDATIONS FOR STRUCTURES:
These shall be as per drawing for structure foundations and as per the Technical specifications for excavation, CC/RCC, Back filling etc. The scope of work is detailed here under.

13.1 **FOUNDATION EXCAVATION:**

Earth excavation has to be done in all types of soils which can be excavated with pick axe & crow bars in all conditions such as dry, wet, slushy etc., and in hard rock requiring blasting including shoring, shuttering and dewatering wherever necessary covering initial lead and lift. Back filling the foundations with excavated earth after laying the foundations.

13.2 **FOUNDATION CONCRETE:**

Laying foundation for main and auxiliary structures, transformer plinths etc., including dewatering the pits before and during the process of laying concrete in the proportion of RCC M20 Nominal mix using 20mm HBG graded metal, PCC M-20 Nominal mix using 20mm HBG graded metal and CC 1:4:8 using 40mm HBG metal as per the approved drawings for structural foundations enclosed to the specification and curing for 14 days including cost of cement, reinforcement steel.

13.3 **BACK FILLING: This item included in the foundation excavation**

Back filling of foundations with excavated soils including watering and consolidating in 600 mm layers and consolidated upto 95% proctor density for normal soils. Providing sand cushion for foundations and flooring area including cost of sand, watering and consolidation upto standard specifications for BC soils.

13.4 **PLASTERING:**

Plastering foundation plinths 12mm thick in (1:3) cement mortar and white washing (two coats) with Janatacem cement paint. All projections of plinths and masonry above ground level shall also be provided with Janatacem cement paint.

14.0 CABLE DUCT:

14.1 **CABLE DUCT IN SWITCHYARD:**

(a) Cable and pipe trenches shall be of reinforced concrete type with removable type reinforcement cement concrete covers. These trenches shall not be used as storm water drains. The bed of the cable/pipe trenches along their length shall be provided a slope of 1:750 to 1:1000 by laying P.C.C (1:2:4) in second stage to drain any rain/seepage water that may enter then. The

overall depth of various type of cable/pipe trenches shall be so decided to take care of above provision of longitudinal slope along their length by laying P.C.C in second stage without compromising the clearance required between bottom most tray of cable trench and bed of the cable trench.

(b) Suitable expansion joints shall be provided in cable/pipe trenches as per IS specification with PVC water stop, bitumen impregnated filler boards, bitumen sealing compound as per specifications or as directed by the purchaser. The water proofing cement additive shall be used for the construction of cable trenches to ensure water tight construction.

(c) The water from cable trenches shall be collected in suitably located sumps from where this water shall be disposed off by pumping into nearest storm water drains. The sump pits shall be of R.C.C construction covered at top with provision of manhole, rungs and installing 2 Nos. of 100% capacity submersible type sump pumps for pumping out the water from the sump pit with auto start and auto shutdown depending upon the level of water in the sump pit. The scope shall also cover all accessories like suction and discharge pipes for transferring water from sump pits to nearest storm water drains. The grade of concrete for the construction of sump pits is M-20 conforming to IS: 456-2000 and this specification.

(d) All cable trenches shall be provided with edge protection angles of size not less than 45x45x5. The pre-cast R.C.C covers for cable trenches shall also have edge protection on all four corners. The length of the edge protection angle shall be decided during detailed engineering. The suitable lugs of reinforcement steel (10 mm dia minimum) shall be welded to edge protection angles for ensuring their rigidity in the concrete. The weight of a single pre-cast R.C.C cover shall not be more than 75 Kg. Suitable lifting hooks of size 10mm rod minimum shall be provided for easy handling of these covers.

(e) The covers of cable trenches shall be of such designs not to allow any rainwater to enter into them. Necessary embedded steel plates of size 100x10 mm thick shall be provided for supporting the cable trays. (f) The top of cable trenches shall be kept 150 mm above finished ground level. The backfilling shall be one with coarse sand. (g) In cable/pipe trenches, construction joints shall be provided between the base slab and sidewall with PVC water stops

14.2 **CABLE DUCT AT ROAD CROSSINGS:**

Cable duct of width 1150mm and 900mm depth by providing 300mm dia RCC Hume pipe (plain ended pipe)& collars required confirming to BIS 458/1988 NP2 class as per drawing in two rows duly embedded in cement concrete (1:2:4) with 50% of 40mm HBG metal and 50% of 20 mm HBG metal shall be constructed as per approved drawing. Rate quoted shall include cost of cement, RCC Hume pipes and other materials and labour charges. Length of duct shall be constructed as directed by the Engineer in charge.

14.3 **CABLE DUCT IN CONTROL ROOM:**

RCC Cable duct is to be constructed as per the approved drawing and as per instructions of engineer in charge at site. G.I. Weld mesh cable trays of 300mm width and 3 meters long with 25mmx25mmx4mm angle frames to be procured by contractor and placed in the duct. These cable trays are to be supported on

Galvanized M.S. Angles of 45 x 45 x 5mm to be procured by the contractor, and fixed in the cable duct as per the drawing. Cable duct will be covered with 7 mm Galvanized chequred plate to be supplied by the contractor. The main cable trenches entering at control room should be sealed for not to enter any creatures.

15.0 PROCUREMENT OF STEEL, FABRICATION, GALVANISATION AND SUPPLY OF STEEL STRUCTURES:

Procurement of steel such as MS angles, plates, Channels and M.S Rounds required for fabrication of structures is with in the scope of contract. The structures shall be fabricated as per the structural drawings and galvanized. The Bidders are required to fabricate proto type structure and submit shop drawings for all the structures, stub setting templates. Mass fabrication of the structures shall be done after their approval. The structural drawings & bill of material will be issued by the Board. The technical specifications for procurement of steel. fabrication and galvanisation of structures are given in the clause –9 of part-B of technical specification. Galvanised bolts, nuts, pack and spring washers required for the assembly of the fabricated pieces to form into a composite structure of standard quality shall be supplied by the contractor. The 'U' bolts for Bus stringing and foundation bolts for main and/or auxiliary structures are also to be fabricated and galvanised by the contractor. The Galvanised nuts required for these bolts are also to be procured and supplied by the contractor.

16.0 ERECTION OF STRUCTURES :

The Structures shall be erected by piece-meal method on the foundations, after allowing the required curing time for the foundations. The members shall not be strained or bent during the course of erection. Care shall be taken to see that the jointing surfaces are clean and free from dirt or grit and fit properly. The structures shall be erected strictly in accordance with the approved drawings. After erection of structures the bolts shall be checked to ascertain that all nuts are fully tight. The contractor shall ensure that none of the bolts are left out. The structures must be truly vertical after erection and no straining will be permitted to bring them to vertical position. The tolerance allowed for verticality is 1 in 360 structure height.

17.0 STRINGING OF BUSBAR & GROUND WIRE :

17.1 The 220 kV bus will be formed with single/Twin/ Quadruple Moose/Zebra ACSR conductor and 132 kV & 33 kV buses will be formed with single/Twin Zebra ACSR conductor and shall be strung to required tension as per site Engineer's direction and as per switchyard layout drawing.

17.2 The stringing of bus conductors covers installation of tension insulator strings, suspension insulator strings, post type insulator stacks and other accessories alongwith tensioning of conductor and clamps. The jumpering to various equipment and the ground wire stringing are also under the scope of this work. The G.I. Ground wire shall be strung as detailed in the layout drawing.

17.3 The insulator strings shall be assembled on the ground. These shall be cleaned, and examined before hoisting. All the parts including current carrying parts shall be clean without grease, paint and dirt. Insulators with cracks or chips or those having glazing defects exceeding 0.5 Sq. Cm shall not be used.

17.4 The sag of bus conductor shall not be more than 1%. Damaged conductors shall not be used. No joints are to be made. The conductor surface shall be clean and smooth without any projections, sharp points, cuts or abrasions, etc., and the conductor shall be continuous in span.

18.0 EARTHING SYSTEM:

18.1 The M.S.Flat of sizes 100x16mm and 50x8mm confirming to IS 2062 shall be procured by the successful bidder. The earth mat shall be as per the drawing No. SETSS 1/99 and extend over the entire switchyard. The earth mat shall be formed with the steel flats buried in the ground at a depth of 500 mm on edge. G.I. wire shall be used to connect the overhead ground wires to the ground mat, along the structures at all the locations.

18.2 All the junctions of the steel flats while forming the earth mat and taking risers from the earth mat for giving earth connections to equipment, steel structures, conduits cable sheaths shall be properly welded. All joints shall be provided with suitable angle pieces for proper contact between Two flats

18.3 Provisions shall be made for thermal expansion of the steel flats by giving smooth circular bends, as per the sketch shown in the earth mat layout. Bending shall not cause any fatigue in the material of bends.

18.4 The earth mat shall be formed by welding 50 x 8 mm steel flat to the 100 x 16 mm peripheral earth conductor. The grounding grid shall be spaced about 5 meters, in longitude and about 5 meters in the transverse directions and the contractor shall be guided by the same and the advice of the Engineer. After completion of earth mat, the earth resistivity may be measured. In case the earth resistance is more than one ohm, earth mat may be extended extra electrodes is installed so that the earth resistance is less than one Ohm.

18.5 All fence corner posts and gate posts shall be connected to the ground by providing 32 mm dia M.S. Rods of 3 meter length near the posts and connected to the main grounding mat.

18.6 All paint, enamel and scale shall be removed from surface of contact in metal surface before making ground connection.

18.7 Earth flat is to be aligned along the main structure supporting structure for the equipment. The risers taken along the main switchyard structures and equipment structures (upto their top) shall be clamped to the structures at an interval of not more than one metre. <u>The risers over the ground level shall be galvanised.All other items of steel coming over the ground and exposed to atmosphere shall also be galvanised. Welded joints shall be painted with zinc rich epoxy paint the cost towards this shall be included in the respective main items.</u>

18.8 50 x 8 mm (flat) ground conductor shall be run in cable routes and shall be connected to the ground mat at an interval of 10 metres.

18.9 Grounding electrodes of 32mm dia 3mtr. Long M.S. rods shall be provided at the peripheral corners of the earth mat. The grounding rods shall be driven into the ground and their tops shall be welded to a clamp and the clamp together with the grounding rods shall be welded to the ground mat.

18.10 Lightning arrestors shall be provided with earth pits near them for earthing (as per the standard practice). Cast iron pipes 125mm in dia, 2.75 metres long and 9.5 mm thick weighing 94.84 kg and confirming to IS:7181/1974 shall be buried vertically in the pits and finely broken coke, shall be filled atleast 150 mm around the pipe for the entire depth. Where it is not possible to go to a depth of 2.75 metres,1.3 x 1.3 M M.S plates, 25mm thick shall be buried vertically in pit of 2 metres depth and surrounded by finely broken coke atleast 2 metre away from any building or structure foundation. The plates shall be atleast 15 metres apart. These earth pits in turn shall be connected to the earth mat. Earth pits are to be provided with RCC collars of 750 mm outer diameter, 50 mm thick and 600 mm height.

18.11 The joints and tap offs when welding is done shall be given a coat of Bitumen paints and then covered with hessain tape to avoid rusting.

19.0 LAYING OF CABLES:

19.1 This section covers laying of cables, in cable trenches, cable ducts in switch yard, control room (two storied building) etc, cable jointing, termination at both ends and testing of the complete cable installation. At the existing substations where there is no cable duct in the switchyard the cables shall be buried in the sand in the ground after digging trenches of 300mm deep,300mm width and these trenches shall be filled with sand. The trench shall be levelled to the formation level of the yard. The cable route length is indicated in the annexure to the schedule of quantities. The cable route is to be marked at regular intervals of 10M by providing cable route marking pegs.

19.2 The control cables will be copper wires of 2.5 sq.mm with number of cores being 2,4,6,10 or 12. Aluminum cables (2,4,6 cores) of 4 sq.mm shall be used as per the directions. Three and half core aluminum power cable shall be used for yard lighting and A.C. supply to equipment. Co-axial cable supplied by the employer for PLCC has to be laid.

19.3 The cable schedules will be provided to the contractor by the Engineer. The cable laying shall be done strictly in accordance with these schedules The length of each cable issued shall be judiciously cut minimizing the wastage as per approval of Engineer.

19.4 All the electric equipment is to be set in place and aligned. Contractor shall provide power and auxiliary connection to the equipment and shall work in cooperation with suppliers representative in obtaining correct direction of rotation and commissioning of the equipment. Cable laying shall also include termination of cables i.e. at both ends of cable of equipment switchgear, control and protection panels, etc., as well as equipment to marshalling boxes and marshalling boxes to switcher panels.

19.5 Secondary terminals of CTs/CVTs/PTs shall be connected to marshalling boxes through cables and then further connected up by cables panels or other equipment.

19.6 Cable lugs shall be provided by the contractor. Dressing/bunching of all individual cores of the cables shall be done in a neat fashion. The contractor shall drill holes

suitable for the cables in cable gland plate.

19.7 Cable lugs shall be compressed on the conductor ends by means of tools. Insulating sleeves shall be furnished and covered over the bare ends of the connections so as to prevent accidental contact with the ground or with the adjacent terminals.

19.8 The insulating sleeve shall be fire resistant, and long enough to over pass the conductor insulation and shall be of correct size of the conductor used.

19.9 Cables entering the control room from out door areas shall be sealed. Spare glands shall be sealed for control panels and some marshalling boxes.

19.10 Standard cable grips and seals shall be utilised for cable pulling after pulling cable, the contractor shall put and attach aluminum cable markers at both ends of the cables and at the control room entry. The cable number and other data shall be punched and the cable markers are to be securely attached to the cables.

19.11 Normally cable shall not be jointed for deviation, the approval of the Engineer shall be taken.

19.12 Sharp bending and kinking of cables shall be avoided.

19.13 Cables shall cross control room basement wall in 4" R.C.C. Hume Pipe embedded in cement concrete in basement wall. Supply of 4" R.C.C. Hume Pipes is in the scope of contract.

19.14 In each cable run some extra length shall be kept at a suitable point to enable one or two straight through joints to be made, should the cable develop fault at a later date.

19.15 The contractor shall pull sufficient lengths of each cable to permit neat arrangement of all cables. Cables should be extend above the glands for 10 to 15cms and identification should be tied up to the cable. Cables coming out from marshalling boxes should be supported by mesh tray and tied up with nylon ropes.

19.16 Cable splices shall not be permitted except where called for or where permitted by the purchaser.

19.17 At cable terminal points where the conductor and cable insulation will be terminated, termination shall be made in a neat, workman like and approved manner by men specialised in this class of work.

19.18 Before any cable terminal connections are made, conductor's insulation shall be pulled out, at the end and identifying ferrules shall be fixed according to the wiring diagrams. Connection shall be made according to the wiring diagram to be supplied by the purchaser.

19.19 Polarity, or phasing shall be checked before connections are made and corrections of polarity, phasing or rotation shall be made by the contractor without additional cost.

19.20 Control cable terminations shall be made in accordance with wiring diagram using colour codes established by the purchaser for the various control circuits, by code marked wiring diagrams furnished to the contractor for this purpose or any other approved means of identification. It is the intent that the contractor shall terminate the cable which he installs.

19.21 Additional work of testing and reconnecting where leads have been brought by the contractor to the terminal boards and connected, but where on further testing, reversed or other rearrangement of load turns out to be necessary shall be performed by the contractor without additional cost.

19.22 Any cable cut short shall be replaced and installed by the contractor at his expense.

19.23 Jointing of cables shall be in accordance with Indian Standard codes. The manufacturers special instructions on materials and tools required for cable jointing work shall be arranged by the contractor.

19.24 Metal sheath and armour of the cable shall be binded to the earthing system of the station by a steel strip wire. Bending the metal sheaths of single core cables in close **threefold** formation shall be as stipulated in the relevant codes of practices.

19.25 The contractor shall furnish two sets of marked up cable layout drawings after installation, indicating altered cable routes and location of straight joints.

19.26 To give an idea of the work involved one set of cable schedules is enclosed

20.0 LIGHTING SYSTEM

The equipment such as lighting distribution boards, lighting panels for normal and emergency lighting, lighting fixtures, luminaries, lamps, wiring accessories junction boxes conduits, conduit accessories, fittings etc. and materials as would be found necessary for illuminating the switchyard with S.V lamps, rechargeable fitting shall be supplied by the successful Bidder as per the specification here under.

20.1 The high pressure sodium vapour lamps with fittings conforming to relevant IS where applicable with latest revisions of M/s. Crompton Greaves make of Model SSS 251H 250 Watts capacity or equivalent of Phillips or GEC or Bajaj make for yard lighting of EHT sub stations are to be offered. The luminary shall have deep drawn anodised single piece aluminum body with circular/oval lamp compartment with suitable control gear consisting of low loss heavy duty electronic ballast like OPAL make, power factor improved capacitor, external electronic ignitor and also provided with detachable parabolic reflectors. The certificates conforming to relevant IS should be submitted and approval obtained before dispatch of the light fittings. These fittings shall be provided for A.C lighting as per lighting arrangement specified in the drawing. The locations of the light fittings in the switchyard will be indicated by the Engineer. The twin tube rechargeable emergency light fittings shall be portable and of approved quality.

20.2 Cables and LTAC panel required for AC supply will be made available by the employer to the contractor. The other materials and labour for giving AC supply

has to be arranged by the contractor.

20.3 The lighting system shall consist of normal AC lighting from 3 phase 415 volts supply.

20.4 The lighting pillar box shall be erected in the switchyard by the successful bidder. The box shall be grouted in C.C. 1:4:8 with 40 mm HBG metal with foundation bolts. Foundation bolts required for the work shall be provided by the contractor himself.

20.5 SUPPORTS FOR FIXTURES:

i) The lights shall be mounted on structures with necessary cantilever arrangement to be arranged by the contractor. All the accessories G.I. Pipes, clamps, etc., required for mounting the fixtures shall be supplied by the contractor.

ii) The steel mounting arrangements shall be as shown in the drawing and same shall be galvanised. The mounting arrangement shall be earthed at two locations with 8 SWG GI Wire to be supplied by the contractor. Mounting arrangement of fixtures is to be procured by the contractor.

iii) All lighting fixtures shall be mounted in accordance with the standard installation practice.

iv) All lighting fixtures shall be assembled, installed and wired. Splices in lighting circuits shall be avoided. Required size of fuses shall be provided for each light junction boxes. Contractor after completing the connections to all outlets shall try out all portions of the lighting system to ensure satisfactory operation.

21.0 ERECTION OF EQUIPMENT:

The contractor has to make his own arrangements for transport of the materials/equipment as given in the bill of quantities and its Annexure to switchyard/ substation site. He is fully responsible to take delivery of materials/equipment and transport to the works stores at site and erect the same as per the directions of the Engineer. However, the contractor shall provide insurance to cover risks during transit from employers stores, storage and erection at site for all fragile equipment like CBs, CTs, PTs, CVTs, LAs ,insulators and Capacitor Banks etc. In case of materials/ equipment not covered by insurance, the contractor shall be very careful in handling them. However, the contractor shall take utmost care in handling the equipment during transport, storage and erection. If a crane is used in the erection of heavy equipment expenditure incurred towards the same shall be borne by the contractor himself.

On completion of the works, all materials left surplus from those supplied by the employer for erection shall be handed over to the Engineer at Switchyard site. Empty drums of conductor, cable and earth wire shall be returned to the Engineer. The contractor shall provide and make all necessary arrangements for the safety of his staff and labourers, at the site of works. The Employer will not in any way be responsible for any accident, minor or fatal to any person at the site of works or for any damages arising there from during erection and this shall be the contractor's responsibility. The staff insurance charges, if any, shall be borne by

the contractor.

All tools and plant, consumables and any instruments required for erection of the equipment shall be arranged by the contractor himself.

The scope of work shall include taking delivery of equipment from the TLC Stores, handling transport to substation site and erection. The contractor shall move all the equipment covered under this section to the proper location, unload and shall accurately level, align and anchor the equipment. The contractor will be responsible for the safety of equipment from the time of his taking over the equipment for erection until the Board takes over.

The scope of work shall also include:

a. Writing names and other details on equipment i.e. Marshalling boxes, Marshalling kiosks, Control and Relay panels etc., with the superior quality white /blue/ black enamel paint including cost of brushes, labour charges etc., complete as per the directions of engineer in charge.

b. Painting of RYB Colour on structures, equipment, isolators, LA's etc with superior quality enamel paint including cost of paint, brushes, labour charges etc., complete as per the directions of engineer in charge.

The manufacturer's installation drawings / instructions and recommendations will be furnished to the successful bidder. The instructions shall be correctly followed in handling, setting, test, commissioning and maintenance of all the equipment. In case of doubt as to the correct interpretation of the manufacturer's drawings or instructions, necessary clarifications shall be obtained from the engineer in-charge of the work. The contractor shall be responsible for the proper erection of the equipment and he will be held responsible for any damage consequent to incorrect handling.

For erection of control , relay and LTAC panels and DC distribution Board in control room the channels are required to be embedded in floor and grouting of foundations bolts in these channels is also to be done by the successful bidder. Materials i.e. 100x50mm Galvanised M.S. Channels, cement and other materials required are to be procured by the contractor. Rubber mat of approved quantity is to be provided in the control room in front of the panels of 137 cm width and 15 mm thickness throughout the panel length.

22.0 SUPPLY OF CO2 FIRE EXTINGUISHERS:

The cylinders containing with carbon-di-oxide(CO_2) gas fitted with appropriate valves confirming to IS 3224 are to be supplied by the contractor. The test certificates issued by the Chief Controller of Explosives Department and cylinder manufacturer's certificates for the supplies already made should be enclosed with the bid. For the supplies to be made against the contract, the certificates of the Chief Controller of Explosives Department for the cylinders must be sent before supplies are made. All the statutory requirements for healthy ness of the cylinders as approved by the Department of Explosives must be satisfied.

23.0 CONSTRUCTION OF FIRE PROCTECTION WALL:

Fire protection walls are to be constructed in between 100 MVA Power Transformers as per the approved drawing.

24.0 SPREADING OF HBG METAL:

The work involves spreading of 20 mm HBG metal for 100 mm thick in switch yard by providing suitable PCC gauge blocks of 100mmX100mm size and 100 mm depth at 2mX2m intervals. Kerb wall of size 400x75x300 mm in RCC (1:2:4) shall be buried in the ground upto 100mm deep. The joints shall be filled with CM (1:3) and Kerb wall shall be provided in the switchyard to retain the metal all along the periphery of the metal spreading wherever it is required. The Kerb wall shall be plastered in CM (1:3) and painted with 2 coats of Janata Cem. Before spreading the HBG metal, care shall be taken to remove all the vegetation and roots of the plant. The trimming of the earth surface for perfect level ground should be carried out and the surface should be watered and rammed for consolidation with suitable means and the earth surface should be given chemical treatment for anti-vegetation growth.

Required metal quantity may be stacked at suitable area for easy handling for pre-measurement. Higher size metal may be separated and removed from the stacks. The crushed metal shall be free from dust. HBG metal shall be brought from the approved Quarry as directed by Engineer-in-charge and samples of the HBG metal to be used may be shown to the Engineer in-charge and prior approval of the engineer shall be taken before use. Spreading should be carried out by dumping the metal at lower heights.

25.0 CONSTRUCTION OF CONTROL HOUSE:

25.1 FOUNDATION EXCAVATION:

This work shall be done as per approved drawing for control House and as per the technical specification given in the Appendix-I. In case hard rock is met with, anchor rods shall be provided with 1000mm length of 16mm dia tor steel after drilling 20mm dia hole for a depth of 500mm inside hard rock and 500mm above hard rock for fixing in position with foundation mat. The anchor rod shall be fixed in the hole with CM (1:2) as directed by the Engineer.

25.2 FOUNDATION CONCRETE:

Cement concrete (1:4:8) with 40 mm well graded hard broken metal for foundations 100mm thick including cost and conveyance of all materials, all leads and lifts and curing etc., complete for finished item of work as base concrete for the walls and columns of the structures. The rate quoted includes dewatering, shoring, shuttering etc., required if any during the works and including cost of cement.

25.3 FOUNDATION BASEMENT:

25.3.1 MASONRY:

ii) Random Rubble or un-coursed rubble masonry with hard stone in cement

mortar 1:6 for foundations using hard granite stones and bond stones (0.16 Cum) upto ground level including cost and conveyance of all materials, cement ,labour charges, all leads and lifts, curing etc., complete for finished work. The relevant APSS clauses 601&615 are applicable to this work.

iii) Coursed rubble stone masonry 2nd sort with hard stone in cement mortar (1:6) using hard granite stones and bond stones (0.16 Cum) for basement and wall above G.L., including cost and conveyance of all materials, labour charges, all leads and lifts, curing etc., complete for finished item of work. The rate includes dewatering, shoring and shuttering required if any and including cost of cement. The relevant APSS clauses 601, 602 &615 are applicable to this work.

iv) Pile foundations for BC soils: This should be done as per the approved drawing for pile foundation and as per the Technical Specification. The C.R. Masonry shall be executed over grade beam from ground level to finished floor level.

25.3.2 PLINTH BEAM: As specified in drawing for control house, plinth beam is to be provided.

25.4 SUPER STRUCTURE:

Brick work in cement mortar (1:6) using well burnt 2nd class country bricks for walls in superstructure at ground floor and first floor and parapet wall in brick work/ 50mm RCC pardi wall as the case may be, shall be provided, including scaffolding, cost and conveyance of all materials, all labour charges all leads and lifts, curing etc., including cost of cement. The relevant APSS clauses 501&504 are applicable to this work.

25.5 FILLING FOUNDATION AND BASEMENT:

Back filling of foundations with excavated soils including watering and consolidating in 600 mm layers for normal soils. Providing sand cushion for foundations and flooring area including cost of sand, watering and consolidation for BC soils. Anti-termite and Anti-weed treatment for foundations, basement and flooring shall be provided. Compaction shall be to the standard specification.

25.6 FLOORING:

i) Providing of CC (1:4:8), 100mm thick for flooring using 20mm and 40mm HBG Metal (each 50%), all leads and lifts, all labour charges, curing etc., including cost of cement. The work shall be carried out as per APSS clause No.402.

i) Providing flooring and dadoing with Vitrified floor tiles of Double charged / Ceramic tiles of approved quality of size and thickness with suitable colour border over cement mortar bed of CM (1:8) 12 mm thick over CC bed already laid or RCC roof slab, including neat cement slurry of honey like consistency spread @ 3.3.kgs per sqm & jointed neatly with white cement paste to full depth mixed with pigment of matching shade inclusive of cost and conveyance of all materials, all leads and lifts, curing etc., complete for finished item of work for rooms other than battery room & PLCC room. The

work shall be carried out complying with APSS clauses 701 & 707.

ii) Skirting for height of 150mm with Vitrified floor tiles of Double charged / Ceramic tiles of approved quality of size and thickness with suitable colour shall be laid over a base coat C.M (1:3) 12 mm thick with cement slurry of honey like consistency spread at the rate of 3.30 kgs per sqm and jointed with white cement paste mixed with pigment of matching shade to full depth provided along the periphery. The work shall be carried out complying with APSS clauses 701 & 707.

iii) Plastering in cement mortar (1:3), 12 mm thick and finished smooth to the required slope and thread lined for flooring including curing, cost and conveyance of all materials, all leads and lifts etc., complete. The Battery room flooring, sides, shall be provided with superior quality Acid resistant paint of approved colour two coats over one coat of primer. The acid resistible painting shall be carried out complying to APSS clauses 1201 & 1214 .

iv) Vinyl flooring with 1.5mm thick shall be provided for PLCC room with adhesive of good quality over CM (1:3) 20 mm thick finished surface. The samples of the materials shall be got approved by the Engineer-in-charge of work before commitment of the work.

25.7 SUPPLY AND FIXING OF DOORS AND WINDOWS:

A) Supplying & fixing of Door frame made of roll formed section of 1.25 mm thick CRCA section, size should be 50 x75 mm with 32 mm rebate for doors along with accessories such as M.S. hinges 100mm long etc., as per the technical specification given in Appendix.

B) Supply & fixing 30mm thick solid bond wood block board type with teak veneer on one face and commercial ply on another face for single/double leaf conforming to IS 2202 (part-1) - 1973 suitable for the door frames of size 1.220 m x 2.135 m, 0.915 m x 2.135 m, 0.750 m x 2.135 m .

C) Supply, fabrication and erection of Aluminium anodized sliding windows using Jindal /equivalent Hindalco sections.

(i) **Two Track windows**: Using aluminum anodized sections of Series C Jindal sections and outer frame top horizontals & both verticals of 20694 of size 62 x 29.5 mm x 1.30thick and bottom horizontal - two track frame of 20703 of size 62 x 29.5 mm x1.20 mm thick, Shutter frame top, bottom and verticals of 20529 of size 50 mm x 20 mm x 1.50 mm thick and Weather interlocking frame of 20531 of size 50 x 20 x 1.50 mm thick with plain clear float glass 5 mm thick fixed including supply and fixing aluminum handles of 100 mm for each shutter, nylon rollers assembly and all labour charges for fixing the fixtures with required no.of screws, bolts and nuts and including labour charges for fixing the frame in position, fixing shutter to frame etc. completed for finished item of work

D) **Double Leaf Door (partly glazed & partly panelled):** using Jindal / eqvivalent Hindalco sections of size 101.60mmx44.75mmx2.40mm @

1.834 Kg/m (Jindal Sec 14021) for frame and 47.62mmx44.45mmx2.02mm @ 1.052 Kg/m (Jindal Sec 19569) for shutter verticals, 47.62mmx44.45mmx1.95mm @ 0.974 Kg/m (Jindal Sec 19571) for shutter top, 114.30mmx44.45mmx2.15mm @ 1.824 Kg/m (Jindal Sec 19574) for shutter bottom, 83.50mmx44.45mmx2.40 mm @ 1.679 Kg/m (Jindal Sec 19525) for shutter middle and fitted with 5mm thick plain glass on upper half and 12mm thick both sides prelaminated cement particle board for lower half of the shutters including cost & conveyance of all materials, 1st quality double action floor springs (for inner & outer rotation), central pivots, rubber beading, heavy duty Mortice lock 6/7 levers with PC/CP handles, labour charges & other incidental charges complete as per the drawing and as per the directions of Engineer-in-charge for finished item of work.

E) **Single Leaf Door (partly glazed & partly panelled):** Using Jindal / eqvivalent Hindalco sections and and outer frame top horizontals & both verticals of 14021 of size 101.6 x 44.75 x 2.40 mm, Shutter frame top of size 44.45 mm x 47.62 mm x 1.95 section 19571 , bottom of size 44.45 mm x 114.30 mm x 2.15 section 19574 and verticals of 19569 of size 44.45 mm x 47.62 mm x 2.02 and Middle lock rail of 19525 of size 83.50 x 44.45 x 2.40 mm and fitted with 12mm thick both sides prelaminated cement particle board for shutter including supply and fixing aluminium handles of 100 mm for each shutter, floor springs/ hydraulic door closure assembly and all labour charges for fixing the fixtures with required no.of screws, bolts and nuts and including labour charges for fixing the frame in position, fixing shutter to frame etc. completed for finished item of work

F) **Single Leaf Door (panelled):** Using sections of size 63.50mmx38.10mmx2.00mm @ 1.054 Kg/m (Jindal Sec 14013) for frame and 47.62mmx44.45mmx1.90mm @ 0.911 Kg/m (Jindal Sec 19570) for shutter verticals, 47.62mmx44.45mmx1.95mm @ 0.974 Kg/m (Jindal Sec 19571) for shutter top, 114.30mmx44.45mmx2.15mm @ 1.824 Kg/m (Jindal Sec 19574) for shutter bottom, 83.50mmx44.45mmx1.70mm @ 1.243 Kg/m (Jindal Sec 19535) for shutter middle and fitted with 12mm thick prelaminated cement particle board truly exterior grade both sides laminated (Bison Lam or equivalent) for shutter including cost & conveyance of all materials, glazing clips, rubber beading, automatic door closer (Hyper brand IS 3564), heavy duty Mortice lock 6/7 levers with PC/CP handles, labour charges & other incidental charges etc., complete as per the drawing and as per the directions of Engineer-in-charge for finished item of work.

G) **Providing & Fixing of Scientific Doors** with metal door frames and door shutters made of galvanize steel (base steel as per IS 513 of 0.58 mm thick D quality, galvanized as per IS 277 with Zinc of 120 GSM)coated with Zinc Phosphate Primer to receive any paint on site or finished with Thermosetting Polyurethane paint of Aliphatic Grade providing high levels of scratch resistance and durability/Epoxy polyester powder for powder coating paint thickness 50-60 microns (Dry film thickness) outer frame section of 100 x 58 mm x 1.2 mm thick, shutter section of 0.80 mm thick galvanized steel sheet pressed (roll formed) for 46mm thick fully flush, double skin door shell seam joints at stile edges, in-fill of honeycomb kraft paper used to give the required rigidity and effective acoustic insulation with 6” Tower bolt – 1 No., 6” D handles – 2 Nos., 10” Aldrop – 1 No., Butt Hinges – 3 Nos, Mortise Lock of approved quality – 1 No, frames fixed to the concrete/masonry wall by means of self

expanding screws complete for finished item of work for **Single leaf Door**.

H) **Providing & Fixing of Scientific Doors** with metal door frames and door shutters made of galvanize steel (base steel as per IS 513 of 0.58 mm thick D quality, galvanized as per IS 277 with Zinc of 120 GSM). coated with Zinc Phosphate Primer to receive any paint on site or finished with Thermosetting Polyurethane paint of Aliphatic Grade providing high levels of scratch resistance and durability/Epoxy polyester powder for powder coating paint thickness 50-60 microns (Dry film thickness) outer frame section of 100 x 58 mm x 1.2 mm thick, shutter section of 0.80 mm thick galvanized steel sheet pressed (roll formed) for 46mm thick fully flush, double skin door shell seam joints at stile edges, in-fill of honeycomb kraft paper used to give the required rigidity and effective acoustic insulation with 6" Tower bolt - 2 Nos., 6" D handles - 2 Nos., 10" Aldrop - 1 No., Butt Hinges - 6 Nos, Mortise Lock of approved quality - 1 No, frames fixed to the concrete/masonry wall by means of self expanding screws complete for finished item of work. **for Double leaf Door**.

I) **Safety Grills**: Using 7.50mm thick Aluminium grill (as approved by the department) @ 3.58 Kg/Sqm and 'U' section of Jindal / Equivalent Hindalco size12.50mmx12.50mmx1.00mm @ 0.133 Kg/m (Jindal sec 17533) for frame including cost & conveyane of all materials, labour charges and other incidental charges etc., complete as per the drawing and as per the directions of Engineer-in-charge for finished item of work.

J) **Ventilator:** Using section of Jindal / Equivalent Hindalco section of size 81.25mmx38.10mmx1.75mm @ 1.093 Kg/m (Jindal Sec 14071) for frame and 'U' section of size 12.50mmx12.50mmx1.00mm @ 0.133 Kg/m (Jindal Sec 17533) for louvers and inserting 5mm thick plain glass panes including cost & conveyance of all materials, labour charges and other incidental charges complete as per the drawing and as per the directions of Engineer-in-charge for finished item of work.

The Technical specifications for Doors and windows are given in the Part-II B.

25.8 BED BLOCKS UNDER LINTELS:

Providing bed blocks in CC 1:3:6 using 12mm to 20mm size HBG metal under lintels with necessary form work including cost and conveyance of all material, all leads and lifts and in all heights and curing etc., for finished item of work including cost of cement.

25.9 LINTELS:

RCC M20 nominal mix (conforming to I.S: 456) / as per approved drawings for lintels using 12 mm to 20 mm size HBG metal conforming to IS:383, including scaffolding, centering machine-mixing, Vibrating curing, finishing, cost and conveyance of all materials, all leads and lifts in all position of the building at all heights, including cost of cement and steel for finished item of work.

25.10 COLUMNS & BEAMS:

RCC M20 nominal mix (conforming to I.S: 456) / as per approved drawings for footings, plinth beams, columns and beams with 12 mm to 20 mm size hard

broken Granite metal conforming to IS:383, including cost of cement , steel centering, scaffolding, shuttering , machine-mixing, Vibrating, curing and finishing to the required tolerances. The work shall be carried out complying with APSS clauses 313,402 & 403.

25.11 SUNSHADES:

RCC M20 nominal mix (conforming to I.S: 456) / as per approved drawings for sunshade of 75mm average thick, using 12mm to 20 mm size hard broken Granite metal conforming to IS:383, including scaffolding, centering, machine mixing, vibrating, curing and finishing to the required slopes and providing lip at the bottom ends including cement plastering 12mm thick with CM 1:5 proportion dubara sponge finish to the exposed faces of sun shades, including the cost and conveyance of all materials

25.12 RCC ROOF SLAB:

RCC M20 nominal mix (conforming to I.S: 456) using 12mm to 20 mm size hard broken Granite metal conforming to IS:383 / as per approved drawings, for required thickness including cost of cement, centering charges, machine mixing, laying, vibrating, curing etc. PVC rain water pipes of 110 mm dia (4Kg/cm2) shall be provided from roof level to ground level. Opening shall be left over for providing 110 dia PVC Pipes while casting the slab.

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SAMPLING AND STRENGTH TEST OF CONCRETE MIX:

The contractor shall arrange for sampling and testing. Samples from each fresh concrete shall be taken as per ISI:1199 and cubes shall be made, cured and tested at 28 days in accordance with IS:516. In order to get a relatively quicker idea of the quality of concrete, optional tests at 7 days may be carried out. The sampling should be spread over the entire period of concreting and cover all mixing units.

Frequency:

Quantity of concrete In work cum.	Number of samples
1 to 5 cum	1
6 to 15 cum	2
16 to 30 cum	3
31 to 50 cum	4
51 and above	4 + one additional sample for each additional 50 cum or part there of.

Three test specimens shall be made for each sample for testing at 28 days. Additional samples may be required for various purposes such as to determine strength of concrete at 7 days etc.

25.13 PLASTERING ITEMS:

i) Plastering to walls in superstructure with cement mortar (1:5) proportion 12mm thick on even surface side and 20 mm thick on uneven surface side with 16mm thick in CM(1:6) for Ist coat & 4mm thick in CM(1:4) for second coat including scaffolding, cost and conveyance of all materials, all leads and lifts and curing etc., complete. Plastering to the roof slab with cement

mortar (1:3) 20mm thick and with Acco proof cement compound of 1 Kg per bag of cement including cost of cement, duly making necessary slopes to drain the rain water. Plastering to roof ceiling with C.M. (1:3) 12mm thick.

ii) Pointing to RR/CRS masonry in basement in cement mortar with CM (1:5) including cost and conveyance of all materials, all leads and lifts and curing etc., complete for finished item of work

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25.14 PAINTING ITEMS:

Cost of tools and plant, paint, Brushes etc., required for the work shall be borne by contractor. The work shall be carried out complying with APSS clauses 908,911,1201,1212,1214,1216 & 1221.

i) Painting to wood work 2 coats with synthetic enamel paint first grade of approved make and colour over a primary coat (total 3 coats including cost of paints and primer etc.)

ii) Painting 2 coats to the steel work (Grills) with synthetic enamel paint first grade of approved make and colour over a primary coat (total 3 coats including the cost of the paint and primer.)

iii) Painting 2 coats with rubber based acid resisting enamel paint of approved quality and make over one coat of primer to the walls , flooring and ceiling of the battery room including the cost of the paint.

iv) Painting the outside of the walls with 2 coats of acrylic exterior emulsion paint having VOC (Volatile Organic Compound) content less than 50 grams/ liter (Snowcem ,Berger or J&N paint) over a primary coat (total 3 coats) in all heights including cost and conveyance of all materials, all leads and lifts, scaffolding and curing etc., complete.

v) Painting the internal new walls with 2 coats of acrylic interior emulsion paint having VOC (Volatile Organic Compound) content less than 50 grams/ liter (Asian ,Berger , ICI , Nerolac , Dulux , Jonson&Nicholson make) over one coat of cement primer paint (total 3 coats). The rate quoted shall include scaffolding, cost and conveyance of all materials, all leads and lifts complete.

vi) Painting and writing names for identification of rooms in Control room/ House with superior quality synthetic enamel paint 2 coats neatly, including cost of paint, brushes, labour charges etc in complete shape as directed by the engineer- in- charge.

25.15 EXHAUST FANS: -

The work involves supply and fixing of 18" (450mm) sweep heavy duty exhaust fans (Khaitan or Equivalent make) fixed with 6mm ply wood sheet in position by providing the frame and making the opening in the masonry and finishing the

same with cement mortar with sponge finish including cost and conveyance of all materials like bolts, nuts, etc., all leads and lifts complete.

25.16 INTERNAL WATER SUPPLY AND SANITARY ARRANGEMENTS.

The porcelain materials such as water closets, wash hand basins and flat back urinal of approved make (Hindware, Neycer, Parryware or Johnson make). Pillar taps, bib cocks and peat valves, etc. are to be supplied by the contractor and work executed as per the directions of the Engineer- in -charge. Samples of these materials shall be got approved by the Engineer before fixing. The technical specification for Sanitary fittings is given in the Clause-11 of Part-II B. Permanent storm water drainage system as given in the Clause-16.2 of Part IIB shall be provided and is in the scope of the contract. G.I. Pipes of 'B' class (Tata, Zenith or Jindal make) 15mm dia at all levels shall be fixed inside the control room and 20mm for external portion from overhead tank to inside the building. 25mm dia G.I. Pipe shall be laid from bore well to overhead tank.

25.17 FALSE CEILING :

Providing and fixing of false ceiling in true horizontal level 600 mm x 600 mm, 15 mm thick Mineral Fibre sheet 600 x 600 of fissured-ANF (Board / Tegular) Edge tiles with a Humidity Resistance of 90% RH, Average NRC 0.50, Light Reflectance >80%, Thermal Conductivity λ = 0.052 to 0.057 w/mk, Fire Performance Class 0/Class 1 using hot dipped Galvanised Steel section exposed surface with pre-coated capping, main Tee of size 24 x 32 mm at every 1200 mm c/c maximum and rotary stiched cross tee of size 24 x 27 mm at evry 600 mm c/c and sub-cross tee of size 24 mm x 25 mm at 1200 mm c/c and wall angle of size 19 x 19 mm fixed to periphery of the wall and the above grid is suspended at every 1200 mm c/c in both directions using 2.0 mm thick pre-straightened GI Wire laying fine fissured butt edge ceiling tiles of 15 mm thick mineral fiber Board including Cost &conveyance of all materials and labour charges such as cutting , fixing of standing of frame work exposing roof making complete as per the directions of the Engineer in charge for the finished item of work. (Recommended brands: ARMSTRONG, Insula etc).

25.18 ENVIRONMENT ASPECTS FOR SUBSTATION :

I. The contractor shall carry out the ground cleaning operations during starting of the construction of substation by manual or mechanical methods only. No chemical method shall be used that lead to chemical contamination of the site.

II. The contractor shall be fully acquainted with the environmental mitigative measures and shall implement as per the directions of engineer-in-charge of the work.

26.0 EXTERNAL WATER SUPPLY ARRANGEMENTS:

Investigating for location of borewell , drilling 165mm dia. borewell ,fixing suitable 3 phase, 3HP motor, and commissioning of the motor, laying pipe line upto overhead tank (1000 liters capacity to be installed) on the roof of control house, plumbing, all the necessary materials etc., complete. The equipment and materials supplied shall be got approved by the Engineer-in-charge before fixing, samples of these materials shall be got approved.

27.0 CONSTRUCTION OF SEPTIC TANK:

Septic tank of size 2 x 0.9 x 2.3 meters (Inner dimension) is to be constructed

with RR walls 450mm wide on all four sides in CM (1:6), CC(1:3:6) with 20 mm HBG metal 230 mm thick at the bottom, providing 75mm thick baffle in RCC 1:2:4 wall as required from the inlet pipe level towards floor of the septic tank and roof slab of 100mm thick in RCC(1:2:4) with 12 to 20 mm HBG metal with an opening of 0.6 x 0.6 meters for manhole. The septic tank is to be provided with 4.5meters long 110 mm dia PVC ventilating pipe with cowl. The outlet from the septic tank should be standard 'T' fitting of glazed earthen wear pipe. The center of the outlet pipe should be about 150mm below the center of the inlet. The floor and sides of the septic tank should be plastered in cement mortor (1:3) 20 mm thick. The thickness of the wall should be 450 mm. The floor of the tank should slope at 1 in 30 towards one end to facilitate cleaning of deposits and the manhole should be above this. The reinforcement for the roof slab and baffle wall should be with 12 mm and 6 mm dia M.S rounds at 150 C/C. The tank should be provided with 0.6 x 0.6 meters size C.I. frame and cover for manhole. The inlet and outlet pipes shall be S.W.G. 101.6mm dia. S.W.G. pipe of 101.6mm dia shall be laid from water closets, wash basin, bath, and urinal to septic tank inclusive of all specials and accessories.

27.1 CONSTRUCTION OF MANHOLES:

Manhole of 457.2mmx 457.2mm size up to 914.4mm depth, to be constructed over bed concrete of 1:4:8 with 40 mm HBG metal 100 mm thick, with230 mm thick brick masonry in CM(1:6), and plastered inside with CM (1:3) 12 mm thick. The channels in the manhole should be semicircular in the bottom half and of diameter equal to seven. Above the horizontal diameter, the sides should be extended vertically to the same level as the crown of the outgoing pipes and the top edge suitably rounded off. The manhole is to be provided with 0.45 x 0.45 meter C.I. frame and cover.

28.0 FORMATION OF ROADS:

WET MIX : Providing, laying, spreading and compacting graded stone aggregate to wet mix macadam specification including premixing the material with water at OMC in mechanical mixer (Pug Mill), carriage of mixed material by tipper to site, laying in uniform layers in sub-base/base course on a well prepared sub-base and compacting with smooth wheel roller of 80 to 100kN / Vibratory Roller 80-100 kN weight to achieve the desired density including lighting, barricading and maintenance of diversion, etc as per Tables 400.11 & 400.12 and Technical Specification Clause 406 MORD / MORTH.

CC ROAD:

Preparation of Surface: The wet mix surface over which CC road is to be laid shall be thoroughly cleaned with brushes and brooms etc. as directed. The base for lying of concrete road shall be free of soft spots and shall be properly drained. The prepared base shall conform to the lines, grades and cross sections as directed. 150 mm thick cement concrete (1:2:4) using 50% of 40 mm HBG metal & 50% of 20 mm HBG metal shall be laid for forming road.

Laying: The use of forms, locations of joints, concrete mixing, transporting placing compaction, curing etc shall be as per clause 1515 of APSS. The joints shall be placed in with ½" thick Mastic pads at the intervals as per the directions of the Engineers- in- charge.

29.0 ERECTION OF GATES:

Main gates shall be supported on RCC pillars. These shall include necessary anchors, rollers, and guides etc. in complete shape to be supplied by the contractor. Main gate pillars shall be constructed as per the approved drawing No. SE/TSS: 2/99. Wicket gate shall be grouted into pillars constructed in brick masonry in CM (1:6). After erection of gates they shall be painted with two coats of superior quality enamel paint over a coat of primer.

30.0 CONSTRUCTION OF COMPOUND WALL:

The compound wall shall be constructed as per the approved drawing No. SE/TSS 1/99. All the material including cement, reinforcement steel and barbed wire shall be procured by the contractor. 10 rows of barbed wire 150 mm apart over the compound wall as indicated in the drawing is to be strung. The M.S. Angles to be grouted in compound wall are to be supplied by the contractor himself and fixed at 4.5 metres intervals. Compound wall shall be painting with two coats white washing with Janata cem on plastered surface. During construction of compound wall due to different levels obtained along the boundary, the contractor may have to raise the basement portion of wall as directed by the Engineer-in-charge. The G.I. barbed wire using IOA 12 Gauge shall confirm to IS: 278 -1978 latest of Glidden type (type-B) with line wire and point wire of 2.5 mm dia.

31.0 FURNITURE FOR CONTROL HOUSE:

Furniture for the control House is to be supplied as per the Appendix-III to the Bill of quantities.

32.0 ELECTRIFICATION OF CONTROL HOUSE:

32.1 Electrification of the Control House is to be done with flexible copper unsheathed PVC cable with No. 14 SWG GI wire for earth continuity in surface PVC box type casing run over the walls of the Control House. Drawing showing the electrification of Control house is enclosed. The various sizes of copper wire to be used for wiring is as follows.

(a) Wiring with 2 of 22/0.30 (1.5 sq.mm) copper wire is to be used for light, bell, fan and exhaust fan.

(b) Wiring with 2 of 56/0.30(4 sq.mm) flexible copper unsheathed PVC cable shall also be used for mains from distribution Board to Lighting points.

(c) Wiring with 4 of 84/0.3(6.0 sq.mm) flexible copper PVC cable in PVC box type casing with No. 8 SWG G.I wire for earth continuity for run of mains from AC panel to 32 Amps ICTP Main switch.

(d) Wiring with 2 of 56/0.3(4.0 sq.mm) flexible copper PVC cable in PVC Box type casing with earthing clips from Main switch to Distribution Board.

(e) Wiring with 2 of 36/0.3(2.5 sq.mm) flexible copper PVC cable in PVC Box type casing with earthing clips from Distribution Board to 15 Amps 3 pin Power plug.

32.2 The work should be carried out in conformity with code of practice for Electrical

wiring and fittings in Government Buildings and Indian Standard Specifications. Samples of fittings, switches and all other materials including wire should be got approved by Engineer in charge before use. Holes made in the Walls for fixtures and other switch Boards etc., should be properly filled with cement mortar and patch work should be painted neatly.

33.0 ENVIRONMENTAL PROTECTION

The contractor shall ensure the following for Environmental Protection.

a) All installations are made in accordance with the Electricity Supply Act, 1948 and IE rules 1956 which will be inspected by the Executive Engineer and clearance given.

b) The contractor shall carryout the ground clearing operation by manual or mechanical methods. No chemical methods that lead to chemical contamination of the site shall be used.

c) The contractor shall provide sufficient drainage arrangements to prevent flooding and erosion of slopes etc., in the site or tower areas.

d) The contractor shall be fully acquainted with the environmental mitigation measures and shall implement as per the directions of the Engineer-in-charge of the project.

e) The contractor shall take necessary precautions while stringing so as to prevent accidents.

f) The contractor shall also ensure during the construction.
   - i) Monitoring of quality of the foundations for the tower pads and transformers in substations.
   - ii) that all safety norms during construction are followed to prevent avoidable accidents during construction.
   - iii) Handling and placing of the equipment to avoid damage of the equipment and accidents.
   - iv) Monitoring whether standard rules are being followed while pulling and tightening the conductors.
   - v) All necessary protection barricades are to be provided to ensure safety and prevention of spreading of cement dust etc.

Contractor ensures that necessary mitigation measures are implemented .

34.0 PROGRESS REPORTS:-

Weekly progress reports showing the actual progress made in the receipt of the materials by the contractor and in completion of various works shall be regularly submitted in duplicate by the contractor to the Engineer. The performance monitoring shall be carried out with the help of computer aided system on MS project and Excel etc.

35.0 RESPONSIBILITY OF THE CONTRACTOR:-

The contractor shall guarantee and be entirely responsible for the execution of the

contract in accordance with the general conditions of contract, specifications, schedules and appendicies. He shall further guarantee and be responsible for proper erection with in the guaranteed completion and performance periods. The contractor should give binding certificate necessarily stating that he has considered supply of all items excluding specified items to be provided by the **APTRANSCO** listed in the Appendix-II for completion of Substation /Bay's works. The contractor should also specify in the binding certificate that he has furnished the additional items with unit rates & approximate quantities required to complete the work and not covered in the schedules of bill of quantities. The lumpsum amount shall be quoted for finished item of work as per Specification of APSS. No extra payment over and above the quoted amount will be entertained. In case any work is not executed based on site condition suitable amount will be deducted and mutually agreed rates.

TECHNICAL SPECIFICATION

PART – II B

CLAUSE NO	TITLE
1.0	Technical specification for common building materials
2.0	Technical specification for excavation
3.0	Technical specification for back filling
4.0	Technical specification for timbering
5.0	Technical specification for cement concrete
6.0	Technical specification for reinforced concrete
7.0	Technical specification for sub-station hardware, clamps & connectors
8.0	Equipment required for O&M of Station Transformer.
9.0	Technical specification for procurement of steel, fabrication and galvanization
10.0	Technical specification for doors and windows
11.0	Technical specification for sanitary fittings
12.0	Technical specification for yard lighting and pillar box
13.0	Technical specification for marshalling boxes for CTs,PTs and CVTs
14.0	Technical specification for marshalling kiosks
15.0	Technical specification for anti weed treatment & anti termite treatment
16.0	Technical specification for permanent storm water drainage system

PART-II B

1.0 TECHNICAL SPECIFICATION FOR COMMON BUILDING MATERIALS

The aggregates such as sand, metal shall be obtained from the source as indicated in lead statements.

1.1 INDIAN STANDARDS:

The aggregates both fine and coarse shall comply with the requirements of I.S.456 for concrete and I.S.383 for sand for use in rendering, plastering and mortar except as hereinafter stated and shall be delivered to and maintained at the site, clean, washed and free from dirt.

1.2 SAMPLES:

Samples shall be submitted to the Engineer and all aggregates used in the work shall be at least equal to the approved sample .

1.3 FINE AGGREGATE:

The fine aggregate for concrete, shall consist of naturally occurring and graded in accordance with the requirements of IS.383 for grading Zone-2 or grading Zone-3 except that no particle shall exceed 3/16 inch. It shall be free from excessive sharpness. Fine aggregate for cement mortar and rendering plastering shall comply with IS 383.

1.4 COURSE AGGREGATE:

Coarse aggregate shall be graded to produce sound concrete and for reinforced concrete shall be such that at least 5% by weight will pass a mesh of a size 1/4 per inch less than the minimum lateral distance between the reinforcing bars or 1/4 inch less than the minimum cover, whichever is smaller.

1.5 WATER FOR CONCRETE:

Clean fresh water shall be used for mixing concrete grout and mortar and curing. The water used for mixing and curing shall be free from deleterious matter and acids and alkaline substances in a solution or suspension. Potable water shall generally be used for mixing and curing concrete.

1.6 WATER FOR DRINKING:

ALL POTABLE WATER SHALL BE FREE FROM DELETORIOUS MATTER AND SHALL BE EQUAL TO OR BETTER than the minimum standard acceptable to the local and state authorities. Filteration and chlorination by approved means shall be installed as necessary to obtain the required standard. Storage and distribution of potable water shall be such as to prevent contamination.

1.7 CEMENT

Ordinary Portland cement of Grade 43 manufactured as per IS 8112 of 1989 shall only be used.

Storage, certification, delivery and testing of the cement shall conform to IS:269, IS:8112 or the latest issue.

IS certification mark is obligatory.

Cement procured from major cement manufacturing plants such as M/s Raasi, M/s Vishnu, M/s Andhra cement co., ACC, CCI, M/s L&T and Coramandel, Priyadarshini, Madras Cements etc will only be accepted for which prior approval shall be obtained from the employer. Copies of invoices and test certificates from the cement manufacturers shall be submitted by the contractor to the Engineer and his clearance obtained before actual use. Such clearance will be provided within a maximum period of one week.

1.8 REINFORCEMENT

Steel required for reinforcement will be procured by the contractor The contractor shall workout the requirement immediately after taking over the site and receipt of working drawings and take the approval of the site Engineer and arrange for procurement

2.0 TECHNICAL SPECIFICATION FOR EXCAVATION:

This clause covers excavation to be done for all the civil works. Excavation rates for all soils for foundations of (i) structures (ii) walls for control room shall be quoted. This rate should be inclusive of dewatering, shoring & shuttering required if any. Sides and bottoms of excavation shall be vertically true. Blasting material for excavation in Hard rock required if any has to be procured by bidder himself. Necessary assistance in the form of approval for procurement of the material will be given by the Board to the extent possible. All rules under explosive act shall be followed by the successful Bidder. Blasting work shall be done only after approval by the site engineer during specified hours. The person in charge of blasting shall satisfy himself that all blasts have exploded before working people are permitted to reapproach work site. Withdrawal of unexploded charge will not be permitted under any circumstances. Unexploded charge shall be flooded with water and the hole marked in a distinguished manner. Another hole shall be drilled at a distance of about 450 mm of old hole in the direction of unexploded charge and fired in usual way. This process shall be continued till the original blast is exploded. Excavation in decomposed or soft rock shall be carried out by crow bars or pic-axes or pneumatic drills etc. If he desires to do blasting, he shall obtain permission of site Engineer and do accordingly.

The rock obtained during excavation is the Board's property and it shall not be used under any circumstances by the contractor. Rock excavated shall be stacked by the successful tenderer at a convenient place within the Substation switchyard as directed by the Engineer-in-charge at no extra cost.

3.0 TECHNICAL SPECIFICATION FOR BACK FILLING:

After completion of foundation, footings and walls and other construction below the elevation of the final grades and prior to back filling, all forms of temporary shoring, timber, etc., shall be removed and the excavation cleaned of all trash, debris and perishable materials. Back filling shall begin only with the approval of the Engineer- in charge. Back filling shall be done with inorganic materials

obtained from the excavation of borrow pits, if suitable, and subject to the approval of the Board's Engineer-in-charge. Back fill shall not be dropped directly upon or against any structure or facility where there is danger of displacement or damage. Backfill shall be placed in horizontal layers not to exceed 60 cm. in thickness. Each layer shall be compacted with proper moisture content and with such equipment as may be required to obtain a density equal to or greater than 95% of maximum as determined by the relevant Indian Standards. Trucks or heavy equipment for depositing or compacting back-fill shall not be used within 1.5m of the foundation/structures, or other facilities which may damage by their weight or operation. The methods of compaction shall be subject to approval of the Engineer-in- charge. Backfill adjacent to pipes shall be hand placed free of stones, concrete, etc., compacted uniformly on both sides of the pipe and where practicable, to a depth of 300 mm over the top of pipes, when tampering around piping care should be taken to avoid unequal pressures.

4.0 TECHNICAL SPECIFICATION FOR TIMBERING

4.1 APPROVAL OF TIMBERING:

The contractor shall submit for the approval of the Engineer, dimensioned drawings showing the methods he proposes to adopt for timbering, the drawings of any temporary staging required for carrying out the works. The Engineer's approval of such drawings shall in no way relieve the contractor of his entire and sole responsibility for safeguarding adjacent works from damage.

4.2 MATERIAL & QUALITY:

All the materials used in the works shall be of the best quality of their respective kinds and specified herein. They shall be obtained from sources and suppliers approved by the Engineer or his representatives, and shall comply strictly with the tests, specified hereinafter or, where tests are not specified in this specification they should conform to the requirements of the latest issue of the relevant Indian Standards hereinafter abbreviated to as (I.S) or other approved National Standards authorised by the Engineer.

4.3 INSPECTION AND TESTING:

All the materials used in the works shall be subject to inspection and tests if required by the Engineer. Unless otherwise stated, the cost of all tests required by this specification shall be deemed to be included in the rates and prices named by the contractor in the bill of quantities. Any material which is prepared or manufactured without notice having been given in writing to the Engineer may be rejected if the Engineer considered that his inspection was necessary during the progress of manufacture of such material.

4.4 APPROVAL:

NO MATERIAL SHALL BE USED IN THE WORKS UNLESS THEY HAVE FIRST BEEN APPROVED BY THE ENGINEER OR HIS REPRESENTATIVE.

5.0 TECHNICAL SPECIFICATION FOR CEMENT CONCRETE:

5.1 CODE OF PRACTICE:

Except where otherwise specified, described or directed all concrete and reinforced concrete work shall be carried out in accordance with Indian Standard 456 Code of practice for plain and reinforced concrete.

5.2 GAUGING CONCRETE:

Aggregate shall be measured in proper gauge boxes. When measuring the fine aggregate due allowance shall be made for the moisture content and the bulk adjusted to suit the mix. The methods adopted for gauging the concrete materials shall have the approval of the Engineer-in-charge.

5.3 CONCRETE PROPORTIONS:

The concrete shall be of nominal mix as specified in the drawings or in the bill of quantities.

5.4 CONSISTENCY:

The quantity of water used shall be sufficient to produce a dense concrete of adequate workability for its purpose, which will surround and properly grip all the reinforcement.

5.5 CRUSHING STRENGTH:

The crushing strength of the concrete should be as specified in IS.456 for the proposed mixes.

5.6 GAUGING WATER:

The General arrangements for the supply of water for mixing concrete shall be to the satisfaction of the Engineer who will determine the quantity of water to be employed in the mix according to the degree of moisture in the aggregate. The quantity of water thus determined shall be accurately measured for each separate mixing in a suitable container.

5.7 MIXING CONCRETE:

Concrete shall be thoroughly mixed to a uniform consistency in the mixing machines of approved types. Mixing shall continue until the cement is thoroughly distributed through out the mass, and shall last at least two minutes or for 80 turns of the mixer after the whole of the water has been added. Any concrete showing signs of initial setting before being deposited shall not be used in the works and shall be removed from the site. The concrete shall be discharged from the mixer on to a level watertight platform or floor or into a water tight receptacle. Normally hand mixing of concrete will not be allowed but where the total quantity of concrete is considerably small, the mixing may be done by hand subject to the approval and entirely at the discretion and satisfaction of the Engineer-in-charge.

5.8 QUALIFIED ENGINEER:

The contractor shall employ qualified Engineer with a qualification of not less than a degree in Engineering who shall be responsible for all concreting carried out for the works. He shall be available at site at all times when concreting is being carried out.

5.9 CONCRETE IN UNSUITABLE WEATHER:

In the event of rain, storms or other severe weather conditions arising, concreting shall be stopped and appropriate temporary stop ends vee groves etc., placed as may be necessary. To meet such circumstances, the contractor shall always have in readiness on the site approved framed sheeting tarpaulins etc., for the protection of newly placed concrete. Should any concrete be damaged due to rain, streams or other weather conditions, the Engineer may order the cutting out /replacement of the damaged concrete at the expenses of the contractor.

5.10 FORM WORK:

This clause is applicable for form work/ centering etc for all the civil works under the specification.

5.10.1 The form work shall conform to clause 10 of I.S.456 latest. The contractor entirely responsible for the sufficiency and efficiency of the form work, which term includes moulds, and also for the safe removal of the same. Before commencing the work he shall submit for the approval of the Engineer, details of the form work he proposes to use but such approval shall in no way relieve him of any of his responsibilities for the sufficiency and efficiency of the work and that it will be resistant to the strains imposed on it in vibrating the concrete and will retain all the fines in the concrete as may be necessary to provide the desired concrete surface.

5.10.2 The form work shall be designed and arranged so that it will not settle under the load and can be stripped and removed without causing any blemish or jar to the concrete.

5.10.3 For beam soffits the contractor shall provide cambers or such other form which will ensure that undersides of beams are truly horizontal or cambered to the extent shown in the drawings.

5.10.4 All forms shall be securely braced and supported to prevent any sagging or bulging during construction. In no circumstances shall wire ties be used. All chamber and radius strips. Liners and cores shall be provided where necessary and shall be due to space and securely fixed. All forms shall be fixed to the proper line and trued up immediately before depositing the concrete. All joints shall be close enough to prevent leakage liquid from the concrete.

5.10.5 Form work for all exposed faces of mass concrete and for all faces of reinforced concrete shall consist of approved material so finished as to produce the concrete surface, finish specified without any loss of fines and without honey-combing or bulges etc., strutting shall be of such design to allow accurate adjustment and easy removal.

5.10.6 Strutting of form work against the sides of the structures which is subject to

movement or vibration will not be permitted.

5.10.7 The inside faces of the forms shall be treated with mould oil or other approved preparation in either case, that will not have deteriorating effect either the strength or prevent the oil or other approved preparation coming into contact with the reinforcement.

5.11 REMOVAL OF FORM WORK:

The length of time between concreting and the removal of the formwork is the sole responsibility of the contractor. It shall, however, be competent, to the Engineer to require a minimum length of time. For structure foundation minimum time limit specified is 24 hours for removing form work from date of placing concrete.

5.12 STOP ENDS :

The position of temporary stop ends for vertical joints shall be as approved by the Engineer. Shuttering to form the stops shall be firmly fixed and secured round the reinforcing bars. Such concrete as passes through the stops shall be hacked off and removed as soon as the concrete has set.

5.13 CONSTRUCTION JOINTS:

Recesses of approved size and type shall be formed in construction joints where required by the Engineer in order to form a key with the following concrete. The cost of all shuttering to construction joints shall be deemed to be included in the rates named in the priced bill of quantities. Before Depositing any concrete resting or abutting on work previously carried out the surfaces and ends of the existing work shall be thoroughly racked to such an extent that no portion of the previous surface remains. Thus roughened surfaces shall be thoroughly cleaned off, brushed and watered immediately before the succeeding operations are commenced. The roughened surfaces shall be coated with cement/sand mortar 1/2 inch thick immediately before the concrete of the next layer is placed (special care shall be taken to put the mortor and fresh concrete thoroughly up against the hardened concrete).

5.14 CLEANLINESS OF FORM WORK:

Before concreting is commenced all formwork shall be scrupulously cleaned and wetted, and the contractor shall adopt all necessary measures to ensure that all debris, dirt, wash water and other refuse is removed. The reinforcement and form work will then be inspected by the Engineer and concreting shall not be commenced until the Engineer gives permission. Such inspection shall not however relieve the contractor any of his responsibility for the correctness of the work in every respect.

5.15 EXECUTION OF CONCRETE WORK:

Section limits: Concreting shall be carried out in Sections, not exceeding the limits specified for particular work.

5.16 CASTING PROGRAMME:

The contractor's casting programme shall be such that such section of work can be satisfactorily completed in one operation after permission has been given to proceed. No claim for overtime working to complete a casting programme will be entertained.

5.17 CONVEYANCE OF CONCRETE:

The concrete shall be conveyed from the mixer to its place in the works as rapidly as possible and in such a manner that there shall be no separation or loss of the ingredients. In no circumstances shall more than half an hour lapse between the time when water is added to the mix and the time when the concrete is finally consolidated in position. The use of concrete distributing chutes at an angle of more than 45degrees from the horizontal will not be permitted without the prior written sanction of the Engineer. In no case shall concrete be dropped from barrows or otherwise from a height of more than one and half meters. The arrangements to be adopted by the contractor for conveying and depositing concrete shall be subject to the approval of the Engineer.

5.18 DEPOSITING CONCRETE:

Before any concrete is put in, the contractor shall carryout any filling of pockets or trimming the sides where found necessary to suit the level and line of the concrete to be laid as directed by the Engineer at site. Unless otherwise approved, concrete shall be placed in one operation to the full thickness of members.

5.19 CONSOLIDATING CONCRETE:

Concrete normally shall be consolidated by means of sufficient number of mechanical vibrators. Hand ramming and tamping will only be allowed where specified or by the prior permission of the Executive Engineer. Hand ramming and tamping where permitted shall be sufficient and efficient such as to produce uniform consolidation.

5.20 The concrete shall be thoroughly worked around reinforcement and against shutters so that all entrained air is duly expelled and the concrete surface when stripped be found to be good and free of the formation of air pockets, honey combing or other defects.

5.21 The concrete shall be worked into position where placed and not allowed to flow, for sloping beams the work of depositing concrete shall start from the lower end and work upwards.

5.22 Except where arrangements, approved by the Engineer are made for placing concrete under water, the areas on which concrete is to be deposited shall be made and kept free from standing water during concreting operations and running water crossing or entering such areas shall be brought under control before concreting is commenced.

5.23 FINISH OF CONCRETE SURFACES:

Immediately any wrought or metal faced for work is struck the surface of the

concrete will be inspected by the Engineer and after any remedial work directed or permitted by the Engineer has been completed to his satisfaction, the contractor shall remove all form marks and other imperfections in order to give uniform appearance. The cost of this work shall be included in the rates for concrete.

5.24 Floor surfaces shall be worked to a smooth even finish to correct levels of falls as indicated in the drawings or as directed.

5.25 Where so directed, concrete floor surfaces shall be treated with silicate of soda grade P.84 diluted with four times its volume of water applied to the work with a watering can or spray and afterwards spread evenly with a mop or brush. Twenty four hours later a second application shall be made and, if any surface still appears porous, further applications of the solution shall be given until the work will absorb no more. Any excess of liquid on the surface after the last coat has been absorbed shall be removed and the surface allowed drying. When dry it shall be washed with the plain water.

5.26 CURING:

The curing period shall commence immediately after the concrete is finally trowelled or secreeded and continue for a period of 21 days. The top and side surfaces of concrete shall be kept moist and be protected from the direct rays of the sun during the period. The contractor shall submit to the Engineer, his proposals for ensuring continuous protection of the concrete during the curing period.

5.27 DEFECTIVE WORK:

CONCRETE WHICH IS DEFECTIVE FROM ANY CAUSE WHAT SO EVER SHALL, IF SO DIRECTED By THE ENGINEER, be cut out and the work reconstructed at the successful bidder's cost. No concrete thus cut out shall be reused.

5.28 The faces of the concrete work shall be sound and solid, free from honey combing. No 'Patching' of any concrete facing will be allowed without the express written permission of the Engineer.

6.0 TECHNICAL SPECIFICATION FOR REINFORCED CONCRETE:

SCOPE:
This section of the specification shall cover furnishing of working drawings with bar-bending schedule and the furnishing, cleaning, bending placing with arrangements for chairs, supports and suitable covers if reinforcement steel plain or deformed for all reinforced concrete works, below and above ground level for complete pits as per drawings and specification.

B) Contractor shall supply, fabricate and place reinforcement to shapes and dimensions, as per IS:2502 "Code of Practice for bending and fixing bars for concrete reinforcement" and as indicated in approved drawings of the contractor.

C) MATERIALS:

Mild steel round bars, cold twisted and deformed bars as medium tensile or high yield strength steel confirming to IS:432 and IS:1786-1979, plain hard drawn steel wire fabric etc will be used as reinforcement as per drawings and directions. Reinforcement accessories shall also be supplied by the contractor in the rates quoted for reinforcement.

D) BAR BENDING SCHEDULES**:**

The contractor shall submit to the Engineer for approval Bar Bending Schedules with working drawings in triplicate, showing clearly the arrangements proposed by the contractor, within one month of receipt of the Letter of Intent or of the receipt of the relevant approved design of the drawings, whichever is later. Upon receipt of the Engineer's final approval of the Bar Bending Schedule and drawings, the contractor shall submit 6 (six) prints of the final drawings with one reproducible print after incorporating necessary modifications or corrections, for final record and distribution. Approval of such detailed drawings by the Engineer shall not relieve the contractor of his responsibility for correctness nor of any of his obligations to meet the other requirements of the contract. Any adjustments in reinforcement to suit field conditions and construction joints other than shown on approved drawings shall be subject to approval of Engineer-in-charge/consultant. The payment for above is deemed to have been included in the rates quoted for reinforcement item. The contractor shall adhere strictly to requirements for a concrete cover over steel reinforcement, projection of bars for bending with future extensions, reinforcement, mesh reinforcement and other items to connection with proper placing.

E) CLEANING**:**

All steel for reinforcement shall be free from loose scales, oil, grease, paint or other harmful matters immediately before placing the concrete.

F) BENDING:

a) Unless otherwise specified, reinforcing steel shall be bent in accordance with the procedure specified in IS: 2502 or as approved by the Engineer. Bends and shapes shall comply strictly with the dimensions corresponding to the approved Bar Bending Schedules. Bar Bending Schedules shall be rechecked by the contractor before any bending is done and he shall be entirely responsible for its correctness. Bars correctly bent shall only be used.

b) No reinforcement shall be bent when already in position in the work, without approval of the Engineer, whether or not it is partially embedded in concrete. Bars shall not be straightened in a manner that will injure the material. Re-bending bars shall be bent by machine or other approved means producing a gradual and even motion. All the bars shall be cold bent unless otherwise approved. Bending hot at a cherry-red heat (not exceeding 845 deg. C) may be allowed under very exceptional circumstances except for bars whose strength depends on cold working. Bars bent hot shall not be cooled by quenching. Bars shall be properly lagged for easy identification.

c) In no case shall the clear distance between bars be less than the diameter of the bar not less than one third of the max. size of coarse aggregate. Reinforcement shall be bent in accordance with procedure specified in IS:2502 code of practice for bending and fixing of bars for concrete reinforcement and shall not be straightened in manner that will injure the material.

d) Expanded portion of reinforcement bars must not be subjected to impact or rough handling and workers will not be permitted to climb on bar extensions until the concrete has sufficient strength so that no movement of the bar in the concrete is possible.

e) Where reinforcement bars are bent at site at construction joints and afterwards bent back into their original positions. Care should be taken to ensure that at no time is the radius of the bend less than 4x bar diameters for plain mild steel or 6 x bar diameters for deformed bars. Care shall also be taken when bending back bars, to ensure that the concrete around the bar is not damaged.

G) SPLICING:

a) All reinforcement shall be furnished in the full length indicated on the drawing. Splicing of the bars except where shown on the drawings will not be permitted without the written permission of the Engineer-in-charge.

b) At a compression splice each pair of lapped bars may be in contact but the minimum clear spacing between the splice and the adjacent splice shall be that specified for adjacent single bars.

H) PLACING IN POSITION:

a) All reinforcements shall be accurately fixed and maintained in position as shown on the drawings by such approved and adequate means like mild steel chairs and/or concrete spacer blocks irrespective of whether such supports are payable or not. Bars intended to be contact at crossing points, shall be securely tied together at all such points by No.20 G annealed soft iron wire or by tack welding in case of Bar larger than 25 mm dia., as may be directed by the Engineer. Binders shall tightly embrace the bars with which they are intended to be in contact and shall be securely held. The vertical distance between successive layers of bars shall be maintained by provision of mild steel spacer bars. They should be spaced such that the main bars do not sag perceptibly between adjacent spacers. Before actual placing, the contractor shall study the drawings thoroughly and inform the Engineer in case he feels that placement of certain bars is not possible due to congestion. In such cases he should not start placing any bar before obtaining clearance from the Engineer.

b) The reinforcement accessories shall be furnished by the contractor. The wire shall be of annealed iron wire quality. Bar supports, charis, spreaders, spacers and over lappings shall be sufficiently heavy to support the reinforcement steel properly and shall be approved by the Engineer-in-charge. No separate payment shall be made on account of the above and the unit rate shall be inclusive of the cost of binding wire, bar supports, chairs, over lappings, spacers etc. However when steel is supplied by the owner/purchaser a detailed record of the above shall be kept for purpose of only calculating the theoretical consumption but not for payment.

I) WELDING:

a) Normal bond laps in reinforcement may be placed by lap or butt-welding reinforcement bars, if asked by the Engineer, under certain conditions. The work should be done with suitable safeguards in accordance with relevant Indian

Standards & IS:2751 for welding of mild steel bars used in reinforced concrete construction as per IS: 2751 and IS:456. Welded mesh fabrics conforming to IS: 1566 may also be used if specified in the Schedule of items and drawings.

b) Field welding of reinforcing bars will not be permitted without the written consent of the Engineer-in-charge. Where welding is permitted it must be at suitable staggered locations. Tests shall be made to prove that joints are of the full strength of bards connected. Welding of reinforcement shall be done in accordance with the recommendation of IS:2751. This clause applies to main and distribution steel only.

c) In those places where welded reinforcement chairs, spacer trusses etc. are necessary like in deep raft etc. and have been detailed in the drawing, any welding necessary for fabrication of such trusses chair etc. is deemed to have been covered under the rates for reinforcement and such spacer, chairs, trusses etc. shall be measured and paid under the relevant reinforcement item. If any steel plates are used as gussets in fabrication of these chairs, trusses the same shall be payable under insert item.

J) CONTROL:

a) The placing of reinforcements shall be completed well in advance of concrete pouring. Immediately before pouring, the reinforcement shall be examined by the Engineer for accuracy of placement and cleanliness. Necessary corrections as directed by him shall be carried out. Laps and anchorage lengths of reinforcing bars shall be in accordance with IS:456, unless otherwise specified. If the bars in a lap are not of the same diameter, the smaller will guide the lap length. The laps shall be staggered as far as practicable and as directed by the Engineer and not more than 50% of the bars shall be lapped at a particular section. Arrangements for placing concrete shall be such that reinforcement in position does not have to bear extra load and get disturbed.

b) COVERS FOR REINFORCEMENT**:**

The cover for concrete over the reinforcements shall be as shown on the approved drawings unless other wise directed by the Engineer. Care should be taken to ensure that projecting ends of ties and other embedded metal do not encroach into the concrete cover. Where concrete blocks are used for ensuring the cover and positioning reinforcement with suitable binding wire, they shall be made of mortar not leaner than 1 (one) part cement to 2 (two) parts sand by volume and cured in a pond for at least 14 (fourteen) days. The type, shape, size and location of the concrete blocks shall be as approved by the Engineer. Their strength shall correspond to the strength of concrete proposed for the structure. Where such cover blocks are used, all cover, spacers shall be secured firmly so that they are not disturbed during vibration.

c) TOLERANCES:

Unless otherwise specified by the Engineer-in-charge, reinforcement shall be placed within the following tolerances as specified in Clause 11.3 of IS:

456-2000:

(a) For effective depth 200 mm or less + 10 mm

(b) For effective depth more that 200 mm + 15 mm

The cover shall in no case be reduced by more than one third of specified cover or 5 mm whichever is less.

K) STORAGE:

Storage	Place of storing	Remarks
A. Cement	Cement to be stored in a place of easy access for proper inspection and identification. It should be placed in a weather-proof building, so that loss due to dampness and other causes and is minimum.	Test of compression strength should be done by the project authorities and result should be intimated to the contractor from time time to time.
B. Reinforcement	Reinforcement procured by the bidder are to be stored diameter-wise in such a place to permit easy identification. The area should be such that water does not accumulate and reinforcement do not get distorted. It should not be stacked directly over ground or near any harmful materials. It should be cleaned of excessive rust before use.	Steel shall not contain contain excess rust or cracks etc.

L) Acceptance Criteria:
Reinforcements shall be checked for cleanliness, proper bending, binding, placing and securing in position with provision for proper cover.

M) Method of Measurements:

a) Bar or any other type of reinforcement used like hard drawn steel wire fabric etc. for reinforced concrete shall be measured by weight in tonnes. The weight shall be arrived at by multiplying the actual or theoretical length measured along standard hooks, cranks, bends, authorized laps etc. whichever is less by the sectional weights. Claims if any for payment for this item shall be submitted with supporting document giving the schedule of bars with sketches. The sectional weight to be adopted shall be IS Sectional weight.

b) Standard hooks, cranks, bends, authorized laps etc. shall be measured.

c) Separator pieces between two or more layers of steel shall not be measured.

d) No payment shall be made for lap welding or butt welding if permitted, Supports, chairs, hangers, etc. of height 300 mm and less, required for keeping the steel in

position unless otherwise specified in the contact. For supporting horizontal reinforcement at heights larger than 300 mm, support drawings shall be prepared by the contractor and payment shall be made for the supports as approved by the Engineer-in-charge, or as actually placed, whichever is less, at the same rate as for reinforcement.

e) No extra payment will be paid for modification of already embedded reinforcement, if required due to faulty fabrication or placement.

f) Dowels neither shown in the drawings nor instructed by the Engineer-in-charge, but required for construction facilities and/or sequences, shall not be measured.

7.0 TECHNICAL SPECIFICATION FOR SUBSTATION HARDWARE, CLAMPS & CONNECTORS

7.1 DESIGN FEATURES AND GENERAL SPECIFICATIONS:

7.1.1 The connectors used for connecting Aluminum bus bars, tubes and ACSR conductors should be made of Aluminum Alloy in conformity with IS: 617/1959 and latest ISS and shall have adequate strength for use in the outdoor conditions. The clamps and connectors shall be in accordance with the requirements in BS: 159/1957 and IS: 5561/1970 or latest ISS.

7.1.2 The clamps and connectors shall be of adequate cross sectional area to be able to carry the full load current as well as the heavy rush of current during short circuits. The connectors shall be rated for 1200 Amps & 2500 Amps rating as per the requirement. The temperature rise shall not exceed that specified in Clause-6 of IS: 5561/1970 when carrying rated normal current of the conductors/tubes connected to the clamps. The clamp shall comprise not more than 2 pieces other than Bolts and Nuts wherever the connection takes place with conductor of ACSR.

7.1.3 The clamps shall be so designed as to be able to withstand the force arising under service conditions due to expansion and contraction.

7.1.4 The clamps shall have current paths as short and direct as possible.

7.1.5 Clevices in which moisture is likely to get accumulated shall be avoided.

7.1.6 The design will be such as to allow the pressure of clamping Bolts and evenly distributed over clamping surface, in order to avoid strain in the conductor and also to avoid hot spots. The contact pressure for which the joint is designed shall be indicated.

7.1.7 Where connections between copper and aluminum are involved suitable bimetallic strips shall be used so that electrolytic action between the two metals prevented. Copper coating on Aluminium will not be accepted.

7.1.8 Bolts, Nuts and Washers used with the connectors shall be made of mild steel and shall be hot dip galvanised and made corrosion resistant as recommended in BS: 1706/1960. Clamps and connectors should be provided with double Bolts, Nuts and Spring Washers.

7.1.9 The busbar clamps and equipment connectors should be so shaped as to facilitate formation of smooth curves for the jumper connections. Where the terminals pad or stud is in vertical plane and the jumper is to be taken horizontally or vice-versa, necessary 'L' clamps should be supplied along with the terminal connectors. The drawings for each type of clamps set should accompany the Tender.

7.1.10 To facilitate easy identification at the time of use of the clamps and connectors, the diameter of the conductors being connected and the terminals of the equipment may be provided on all the clamps and connectors.

7.1.11 The finishing of the clamps shall be neat in appearance and should exhibit high workmanship.

7.2 DESIGN FEATURES AND GENERAL SPECIFICATION FOR HARDWARE FITTINGS

7.2.1 TENSION HARDWARE FITTINGS:

The hardware fittings shall be made of heat treated malleable cast iron or drop forged steel and shall be hot dip galvanised. The clamps and fittings shall be tested in conformity with ISS: 2486,Part-I and II/1963 with latest amendments, if any. The hardware shall be suitable for use with ball and socket type insulators of 16 mm pin diameter.

(a) The tension set hardware include complete set of tension materials without disc insulators) with bolted type tension clamps and metal parts such as 'U' bolts, ball link, socket eye etc., complete in all respects and include all fittings necessary for bus bar stringing.

(b) The 'Tension Clamp' body shall be made of Grade-A malleable cast iron to IS:2108/1962 or Grade-B malleable iron to BS:310/1958. They should ensure long and continued service withstanding the vibration fatigue and free from blow holes and other casting defects such as cracks etc.,

(c) The tension clamps shall be of bolted type and shall not permit slipping of or damage to or failure of the complete conductor at any part thereof at load less than 95% of the ultimate strength of the conductor specified in the Appendix.

(d) The slipping strength and ultimate strength shall not be less than the specified values given below

Sl.No.	Particulars	Suspension Sets(Kgs)	Tension Sets(Kgs)
I	Hardware Clamps for ACSR Conductors :		

	a.	Sliping Strength	3,300	6,650
	b.	Ultimate Strength	7,000	7,000
	c.	Minimum failing load	7,000	7,000
II.		Other Hardware Minimum failing loads	7,000	7,000
III.		Tension Clamp for 7/8" SWG Galvanised steel wire minimum breaking minimum breaking load of standard wire.	6,400	
IV		Fault Current Ratings	(On 220 KV 40 KA for duration of 3.0 Seconds and on 132 KV, 31.5 KA for a duration of 3 seconds and on 33 KV 25 KA for a duration of 3.0 Seconds).	

(e) Bolts, Nuts and Washers used shall be made from mild steel and shall be hot dip galvanised and made corrosion resistant as recommended in BS: 1706/1960. The insulator stringing hardware including clamp body be hot dip galvanised. Electro galvanisation is not acceptable.

(f) The design will be such as to allow the pressure of clamping bolts evenly distributed over clamping surface in order to avoid strain in the conductor and also to avoid hot spots.

(g) The 'U' Bolts shall be fitted with hexagonal nut and a checknut and washers. After tightening, two or three threads must be left over.

(h) The clamps shall be suitable to Panther/ Zebra/Moose ACSR Conductor

(i) All forging and casting shall be of good finish free from flaws and other defects. The edges of the outside fittings such as at the eye, clevis and holes shall be rounded. For use with ACSR conductors, Aluminium liners are to be provided for clamps and Aluminium saddle shall be provided at each 'U' Bolts point for tightening the 'U' bolts.

(j) The edge of the clamps body shall be atleast one and half times the diameter of the bolts to ensure prevention from breakage while tightening.

(k) The clamps shall be so designed, so as to withstand the force arising under service conditions due to expansion and contraction.

(l) The design of the suspension clamps shall be such as to avoid all hard spots and links which are likely to injure the conductors. There shall be no sharp radii of curvature and ridges. Suspension clamps shall be so designed that the effects of vibration, both on the conductor and fittings itself are minimised. The clamps shall have ease of oscillation around a horizontal axis, light in weight, low effective power loss and small moment of inertia. The suspension clamp shall permit the conductor to slip before failure of the conductor occurs and shall have a sufficient slipping strength to resist conductor tension under broken wire conditions. It shall have sufficient contact surface to

minimise damages due to fault currents mentioned in the Appendix. The suspension clamps shall be provided with suitable socket or clevis eye for connecting it to the insulator string.

(m) The body of the suspension clamp shall be made out of the high strength Aluminium Alloy and the hardware fittings with malleable cast iron.

(n) Tension clamps of compression type which require hydraulic compressors will not be accepted.

(o) The 'U' bolt type tension clamps shall be designed in such a way to bond the conductor through an arc of 60 to 120 deg. and to grip the conductors with pressure applied by 'U' Bolts.

7.2.2 BOLTED TYPE TENSION CLAMPS FOR GROUND WIRE:

The tension clamp shall be made of hot dip galvanised malleable iron and shall be suitable for stringing 7/8" SWG ground wire for a working tension of 450 Kgs. The clamp shall be completed with eye link anchor shackles, bolts, nuts ,washers and tinned flexible copper earth bonds. The clamp shall have provision for fixing the copper earth bonds. The slipping strength of the clamp should not be less than 95% of the breaking load of 6400kgs and the breaking strength shall not be less than the ultimate breaking strength (minimum breaking load) of the ground wire.

(a) The tension clamp shall be designed to withstand the following bus tensions:

(i) For 220/132 KV Bus : 900 Kgs. for each conductor.

(ii) For 33 KV Bus : 450 Kgs. for each conductor.

(b) All tests as per ISS shall be conducted as per latest ISS

7.3 The successful contractor should submit manufacturer drawings for each item in triplicate well in advance of commencement of the bus stringing and jumpering works and got approved by Employer. All tests in accordance with latest issue of ISS/BSS shall be performed on all the hardware, clamps & connectors before supply by manufacturer.

8.0 TECHNICAL SPECIFICATION FOR STATION TANSFORMER

8.1 STANDARDS:

Except otherwise specified or implied herein the station transformers shall comply in all respects with the latest applicable IS-2026 and CBIP manual on transformers.

8.2 TYPE AND RATING:

The transformers shall be of core type, three phase, oil immersed and shall be suitable for outdoor service. The transformers are normally used as step down transformers. The required rating electrical characteristics of station Transformers are furnished below:

a)	Type	:	Core type 3 phase oil immersed outdoor type.
b)	Rated KVA	:	250 KVA for 220 KV SS 100 KVA for 132 KV SS
c)	Service	:	Outdoor as step down.
d)	No. of phases in the system	:	Three
e)	Frequency	:	50 Hz.
f)	I) Double wound or Auto Transformers	:	Double wound
	ii) Winding conductor material	:	Copper
	iii) Winding connections	:	a) HV Delta b) LV Star
g)	Type of cooling medium	:	Mineral oil
h)	Rated Voltage	:	
	i) Higher voltage winding	:	33 KV
	ii) Lower voltage winding	:	400 volts.
i)	Highest System Voltage	:	
	i) Higher System Voltage	:	36 KV
	ii) Lower Voltage System	:	As per IS:2026
j)	Method of System Earthing	:	Solidly Earthed Neutral
k)	Insulation level of winding	:	

	i) 33 kV windings	:	170 KV	
	ii) Lower Voltage winding	:	As per IS:2026	
l)	Power Frequency voltage	:		
	i) 33 K V windings	:	70 KV	
	ii) Lower Voltage winding	:	As per IS 2026	
m)	Vector relationship required between windings and vector symbol	:	DY 11	
n)	Tap changing	:	off circuit	
o)	Tapping shall be provided in Higher voltage winding for Variation of LV voltage.	:	0 to –10% in 2.5% steps.	
p)	Type of Cooling	:	ONAN	
q)	Impedance Voltage at 75 Deg. C (with+10% Tolerance as per ISS:2026)	:	250 KVA 5%	100 KVA 4.5%
r)	Whether Neutral Terminals are to be brought out	:		
	i) Higher voltage winding	:	No	
	ii) Lower voltage winding	:	Yes	

The above requirements shall be complied with and particulars furnished along with bid. The No load loss and Copper loss particulars also should be furnished.

8.3 GUARANTEE:

8.3.1 The no load loss and copper loss in watts at rated voltage, frequency and output shall be guaranteed by the Bidder.

For the purpose of comparison of tenders, the quoted prices shall be capitalized for the transformer losses at the following rates per KW.

No load loss	--	Rs.1, 04,424
Copper loss	--	Rs. 41,770

If the guaranteed losses are subject to plus tolerance, the maximum tolerance shall be added to the losses in computing the capitalized costs. The maximum plus tolerance applicable may please be stated clearly in the bid.

For the purpose of penalty computations, the test figures of the losses will be compared with the corresponding guaranteed figures. No tolerance is permitted over the test figures of the losses. The penalties shall be calculated at the following rates.

i)	Penalty for no load loss	:	Rs.2,08,848}{ }{ per
ii)	Penalty for copper loss	:	Rs. 83,540 }{ KW

For fraction of kilowatt, penalties shall be applied prorata.

8.3.2 The temperature rise of the windings, core and oil of the transformer shall not exceed the specified limits in latest IS: 2026 and the guaranteed values shall be furnished in the bid submitted.

8.4 CORE:

8.4.1 The core shall be built up with high-grade non-ageing, low loss and high permeability CRGO Silicon, steel laminations specially suitable for transformer cores. The core shall be provided with lifting lugs suitable for lifting the complete core and coil assembly of the Transformer. Low loss cores like amorphous cores shall be acceptable and the coil shall be so fixed in the tank that shifting will not occur when the transformer is moved or during a short circuit.

8.4.2 The transformers shall be suitable for over fluxing (due to combined effect of the voltage and frequency) upto 12.5% on any tapings without core heating. The maximum flux density in any part of the core under such conditions shall not exceed 1.9 tesla.

8.4.3 WINDING:

(a) The windings shall be so designed that all the coil assemblies of identical voltage ratings shall be interchangeable and field repairs to the windings can be made readily without special equipment.

(b) The windings shall be clamped securely in place so that they will not be displaced or disturbed during short circuits. The assembled core and windings shall be vacuum dried and suitably impregnated before removal from the treating tank. The conductors used in the coil structure shall be best suited to the requirements and all permanent current carrying joints in the windings and the loads shall be welded or brazed.

8.4.4 INSULATING OIL:

The oil for first filling shall be supplied with each transformer. The oil shall comply in all respects with the provisions of latest IS:335.

8.4.5 TANK:

(a) The transformer tank and cover shall be fabricated from good commercial grade low carbon steel suitable for welding and of adequate thickness. The tank and the cover shall be welded construction. The main tank body shall have sufficient strength to withstand without permanent distortion (1) filling by vacuum at

750mm mercury without oil and (2) continuous internal gas pressure of 0.35mm atmospheres upto operating level of oil. The tank cover shall be bolted to the tank and the transformer design shall be such that the tank will not be split between the lower and upper cooler connection for untanking.

(b) The design of the tank, the lifting lugs and boxes shall be such that the complete transformer assembly filled with oil can be lifted with the use of four lugs without any damage or distortions. The tank shall be provided with two suitable copper alloy lugs for the purpose of grounding.

(c) The tank shall have two filter valves, one at the bottom at one end and one at the top at the opposite end.

(d) The transformer tank shall be supported on a structural steel frame with forged steel or cast steel single flanged wheels suitable for moving the transformer completely filled with oil.

8.4.6 TAPS:

(a) Tapings shall be provided in high voltage winding from 0 to –10% to obtain a voltage variation in steps of 2.5% on high voltage side. The transformer shall give full load on all taps.

(b) Tap changing shall be carried out by means of an externally operated self positioning switch and when the transformer is in de-energized condition. Switch position No.1 shall correspond to the maximum plus tapings. Provision shall be made for locking the tapping switch handle in position.

8.4.7 OIL PRESERVING EQUIPMENT:

(a) The conservator shall have a shut off valve and sump with a small drain valve and stop cock.

(b) A silica gel breather shall be provided for each conservator vessel. The external atmosphere shall not be in continuous contact with the Silica gel.

8.4.8 BUSHINGS:

The bushings used shall comply with IS:2099 and the dimensions for the following voltage classes shall conform to latest ISS mentioned against them.

Voltage Classes	Indian standard Specification For porcelain parts	For metal parts
upto 11 KV	IS:3347 part-I Section..1965	IS:3347 Part-I Section..1967
36 KV	IS:3347 part-V	IS:3347 Part-V

8.4.9 CENTRE OF GRAVITY:

The center of gravity of the assembled transformer shall be low and as near the vertical center line is possible. The transformer must be stable with or without oil.

8.4.10 ACCESSORIES:

The supply shall include the following with each transformer.

a) First filling of oil conforming to ISS:335

b) FITTINGS AND ACCESSORIES:

i) Rating and terminal marking plates.

ii) Earth terminals..2 Nos.

iii) Lifting lugs.

iv) Main oil guage.

v) Drain valve with plug or cover plate..20 mm dia.

vi) Thermometer pocket

vii) Air release plug.

viii) Oil conservator with drain plug oil filling hole with blanking flange.

ix) Pressure relief vent.

x) Silicagel breather.

xi) Top filling valve with plug or cover plate (32 mm dia)

xii) Mounting channel and plain rollers.

xiii) Lock with keys in duplicate for externally operated tap switch.

xiv) H.V. Outdoor bushings.

xv) L.V. Outdoor bushings.

xvi) Oil level indicator.

The transformer shall be supplied with all the accessories and fittings for normal operation whether they are mentioned above or not and the transformer shall be complete in itself.

8.4.10.1 The following required equipment for operation & maintenance of station transformer are also to be supplied along with the station transformer and erected.

1. HG fuse set with fuse of required gauge.

2. LT distribution box containing two/three sets for distribution with MCCBs/fuse units and cable of adequate size.

3. Earthing system.

8.4.11 PAINTING:

Before dispatch all steel works not under oil shall be painted with a primary coat of anti-corrosive paint of durable nature and two coats of dark admiral gray paint. The interior surfaces shall be painted according to the manufacturers standard practice.

8.4.12 ASSEMBLY AND TESTING:

(a) The transformer shall be completely assembled and tested at the factory in accordance with IS:2026

(b) Three copies of each test report shall be furnished for approval.

8.4.13 TERMINALS:

H.V. Transformer terminal shall be suitable for connecting Zebra/Deer conductor and L.V. terminals for connecting PVC Aluminum power cable.

8.4.14 CLEARANCE:

The minimum clearance in air shall be as follows (as per CBIP Manual)

Voltage	Medium	Clearance phase to phase in mm		Clearance phase to earth in mm	
		Open	Enclosed	Open	Closed
400V	AIR	40	25	40	20
33 KV	AIR	350	--	320	--

8.4.15 DRAWINGS:

(a) The bidder shall supply four copies of the drawings, which will describe the equipment in detail for approval. After approval six complete sets of final drawings for each substation shall be sent. The reproducible drawings may also be supplied.

(b) The following drawings for each item are to be supplied as a part of the contract.

i) Outline dimensional drawings of transformer and accessories.

ii) Assembly drawings and weight of main components.

iii) Drawings giving the weights for foundation.;

iv) Tap changing and nameplate diagram.

v) Large scale drawings of high and low tension windings of the transformers showing the nature and arrangement of insulation and terminal connections.

vi) Bushing drawing and Specification.

vii) Test reports (for both type and routine tests)

(c) Six copies of instruction booklets/operation and maintenance manuals shall be supplied for each substation. Descriptive literature and data on transformer construction, winding, bushing, heat exchanger etc., shall be supplied for each transformer.

8.5 Equipment required for O & M of station transformer

The following required equipment for operation & maintenance of station transformer are to be supplied and erected.

1. HG fuse set with fuse of required gauge

2. LT distribution box containing three sets for distribution with MCCB units.

3. Earthing system

4. Electronic energy meter of 0.5 class accuracy with 100/5 A CTs fitted in a out door pilfer proof M.S box.

9.0 TECHNICAL SPECIFICATION FOR PROCUREMENT OF STEEL, FABRICATION & GALVANISATION

9.1 STEEL:

Procurement of steel, such as MS angles, plates, channels, RS Joists, MS rounds foundation bolts and earthing electrodes etc, required for fabrication of structures and reinforcement steel for structural foundations are within the scope of contract. The following provisions shall apply in connection with the procurement of steel by the tenderer.

a) Steel used for fabrication of structures shall be of mild steel of tested quality as per IS-2062. The steel rods used for foundation bolts and U-bolts shall be of mild steel of tested quality as per IS-432. Torr steel required for structural foundation shall be of tested quality as per IS-1786.

b) The tenderer shall take into account the fabrication wastage while quoting the rate. The Board will not accept any liability in connection with the wastage of steel during fabrication or otherwise.

c) The substitutions if any of steel sections of the structure parts by higher sizes due to non-availability of or otherwise shall be to the suppliers account. The Board will not accept any liability on this account. This is also applicable in case of steel sections for reinforcement steel for structural foundations.

d) The steel shall be procured exclusively from the main steel producers approved by the employer. However, sections not rolled by main producers can be procured from re-rollers approved by the employer provided that Re-rolling of structural steel sections is done from billets/ingots of tested quality& Re-rolled sections are duly tested as per relevant standards.

e) Bolts, Nuts and Spring washers are under the scope of contract and these shall be procured from approved suppliers of the Employer. The bolts and nuts shall conform to the following Standards.

   i) The bolts shall conform to IS-12427/1988 and its latest revision if any.
   ii) The bolts shall be of product grade C as specified in IS:1367(Part-2) 1979.

iii) The nuts to be used for the above bolts shall conform to the requirements of IS-1363 (Part-3) 1984. The nuts shall be of a property class 5 as specified in IS-1367 (Part-6) 1980. The proof stress value shall be as follows:

Nominal nut size	Proof stress (Sp) N/Sq.mm
M 16	490
M 20	500

The bolts shall be of property class 5.6 as specified in IS-1367 (Part-3) 1979. The bolts and nuts shall be hot dip galvanised in accordance with the requirement of IS-1367 (Part-3) 1983.

f) The spring washers shall be of type B and shall conform to IS-3063-1972, including latest revision if any. The spring washers shall be made from high quality spring steel conforming to IS-4072-1975, including its latest revision if any. The spring washers shall be electro-galvanised conforming to Grade 4 of IS-1573-1986 with a coating thickness of 25 microns.

g) The contractor shall submit copies of invoices/test certificates for the purchase of steel, bolts & nuts, washers etc. to the engineer and obtain his clearance before actual use. Such clearance will be issued within a maximum period of one week.

9.2 FABRICATION & WORKMANSHIP:

In the fabrication of structures, the process of straightening maintaining clearances, cutting holding, assembly, riveting , bolting, welding are to be in accordance with clause 32 to 39 of IS 800. All the structures shall be fabricated in accordance with the drawings supplied by the Board. All the structural members shall be fabricated to bolt together easily on the site without any unique strain on the members or bolts. All steels sections, before any cutting work is started shall be carefully straightened and toned by pressure and not by hammering. Strict quality control shall be maintained at all stages of fabrication to ensure proper performance of structures.

9.3 GALVANISING :

All members of structures and special structures "U" bolts etc., shall be hot dip galvanised. Stubs and foundation bolts shall also be hot dip galvanised upto one third length of stub and foundation bolts from top.

Galvanisation shall be done by hot dip process in full compliance with IS:2629 - 1966 and IS:4759 - 1968 where any materials is to be galvanised it should be free from greases and paint, etc., during and after galvanisation. After galvanising the surface shall be free from all sharp edges and metal nodules and there shall be no clogging of bolt holes due to the stay of zinc in the holes after galvanising.

Tests on the galvanised samples shall be carried out regularly strictly in

accordance with IS:2633 -1972. The entire batch represented by the defective sample piece shall be rejected and re-galvanised to satisfy the test requirements. In addition strict inspection of the galvanised materials shall be carried out by the contractor before dispatch and any visible or suspected defects shall be rectified forthwith.

The bidders shall submit with their bids full specification of the quality of zinc to be used stating its percentage purity and the process of galvanisation adopted by them along with the devices and facilities available with them for more authoritative and better control of quality of galvanising. They shall also state the maximum percentage increase in the weight of the completely galvanised materials compared to their black weight and steel with weight of the galvanised structures.

9.4 MARKING:

Each part of steel structure shall be clearly stamped with 20mm steel stamping die with identification number mark or symbol to facilitate erection corresponding to those shown in erection drawings. These marks shall be applied before painting/galvanising and shall be clearly visible after painting.

9.5 BOLTS, NUTS & WASHERS:

Galvanised bolts and nuts and plain spring washers required for the assembly of the fabricated pieces to form into a composite structure of standard quality shall be supplied by the Contractor. The U-Bolts for stringing and foundation bolts for main and auxiliary structures are however to be fabricated and supplied by the tenderer. The required MS rounds for U-bolts and foundation bolts shall be procured by the tenderer. The fabrication of U-bolts and foundation bolts will be under the fabrication item. The nuts required for these bolts shall be procured by the tenderer.

9.6 PACKING:

The materials shall be packed in bundles securely wrapped four times around at each end with No.9 SWG wire with ends twisted as per the practice of latest ISS.

9.7 INSPECTION & TESTING:

A representative of the purchaser shall have access to the contractor's works at any time during working hours for the purpose of inspecting the manufacture, storage and assembly of steel structures. He may select test samples going into the manufacture and the contractor shall furnish facilities for testing such samples. The check assembly shall be done with the structure in a horizontal position and not erected vertically. For the check assembly, bolts and nuts shall be no more than finger tight.

9.8 WEIGHTS OF STRUCTURES :

The weight of structures as per bills of materials will be considered for the payment of supply, fabrication, galvanisation and erection bills. If the designed steel section is not available, higher section shall be used only after obtaining prior approval from the Board. In case of such approvals the cost of supply, fabrication, galvanisation and erection for extra weight shall be borne by the Bidder.

9.9 RESPONSIBILITY OF THE TENDERER:

The tenderer is responsible for proper workmanship and accuracy of fabrication of the materials. The defective materials are to be rectified or made good at his cost. He is also responsible for safe delivery of the goods in good condition at the destination i.e. substation site.

10.0 TECHNICAL SPECIFICATION FOR DOORS AND WINDOWS

10.1 DESCRIPTION OF THE COLD FORMED M.S.HOLLOW DOOR FRAMES.

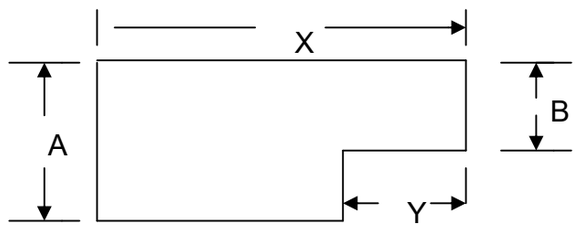
The Cold formed pressed steel frames shall be manufactured from cold rolled sheet metal of 1.25 mm thickness and of size 105 x 60 mm. The frames shall be thoroughly cleaned. Free of rust, mill scale, dirt, oil etc., by mechanical and chemical means before giving an anticorrosive red oxide primer. The selection of raw material and construction of frame shall be as per the recommendations of Indian Specification I.S.513-1963 and I.S. 4351-1976.The door frame shall be provided with 100mm M.S hinges and along with accessories

a) **MATERIAL FINISH**: The door and frames shall be supplied in red oxide primer painted finish.

b) **TOLERANCES:** Tolerances on thickness and other dimensions of the materials shall be as per I.S. 513 and I.S.4351 latest issued.

c) **PACKING**: Door frames shall be packed in knockdown conditions ready to be erected at site. Each frame shall be packed separately along with corresponding accessories in a gunny wrapping.

d) Section dimensions of the M.S cold formed Hollow door frames X=105mm, Y=30mm, A=60mm, B=45mm Thickness of MS steel used shall be 1.25 mm as per I.S.4351-1976.



10.1.1 ACCESSORIES:

The following accessories shall be supplied along with each door frame.

i) Built in (welded) hinges of 100 mm size--3Nos. for single shutter door and 6 Nos. for double shutter door.
ii) Hold fasts -- 6 (six) Nos.
iii) Bottom tie of 10 mm square rod -- 1 (one) No.
iv) Fastering bolts and nuts -- 8(eight)Nos.
v) Stiffners -- 4 Nos on either side and 3 Nos on top at suitable intervals.

10.1.2 SHUTTERS :

Shutter shall be of block board solid core flush type commercial make conforming to I.S. 2202(part - 1) 1973 Thickness shall be 30 mm. Acceptance tests will be done in the presence of Board's Engineer on the doors as per above ISS.

10.1.3 SIZES OF SHUTTERS FOR DOORS.

The over all sizes for doors are mentioned in the schedule. The actual sizes of doors shutters shall be manufactured duly deducting 2 x 75 mm width wise and 1 x 75 mm length wises towards provision of MS door frames (No MS frame for the bottom threshold)

10.1.4 ACCESSORIES :

The following accessories shall be supplied along with each Door.

i)	Tower Bolts	:	3 Nos. for double leaf As per the note.
ii)	Tower bolts	:	2 Nos. for single leaf in specification.
iii)	Al drops	:	1 No.
iv)	Stopper	:	2 Nos. for double leaf.
v)	Stopper	:	1 No. for single
vi)	Handle	:	1 No.

10.1.5 FIXING:

It is the responsibility of the tenderer to fix the shutters with accessories for the doors and windows.

10.2 ALUMINIUM WINDOWS, VENTILATORS AND FITTINGS:

10.2.1 The aluminum windows, ventilators shall conform to relevant IS specifications. All windows, ventilators, louvers etc., shall be of sizes as specified and conform to the description in the respective item of work. Whether or not specifically mentioned, all fixtures and fittings necessary for the satisfactory operation of the windows and ventilators shall be provided. The windows and ventilators shall be obtained from an approved manufacturer fabricated at site or specific approval of such purchase and fabrication shall be obtained before hand. Sample shall also be totally approved before further manufacture starts, unless this is waived in writing by the engineer. The contractor shall obtain windows with friction hinges in place of windows with pigsties if so directed by the engineer. For centre hung and top hinge ventilators suitable spring catch/pulley and chord arrangement shall be provided for facility of opening. Whenever fly mesh over windows have been called for, they shall be fixed on the window and suitable lever type or roto type arrangement shall be provided for opening or closing of the glazed panels from inside. Prior approval of the engineer shall be taken before order is placed with manufacturer. Aluminum safety grills shall be provided as directed by Engineer-in-charge.

10.2.2 Where specified, aluminum windows supplied shall be air tight, for this purpose, the contractor shall provide necessary padding material such as rubber, felt or any other approved material.

10.2.3 Glazed windows, louvers and ventilators shall be provided with either clear or pin headed glass 5 mm thick (as may be directed during execution), which shall be free from all blemishes and shall conform to IS: 1761. It should be clearly understood that glass which does not have uniform refractive index or which is wavy, will be rejected.

10.2.4 The rate quoted shall be inclusive of glazing free from all blemishes. The workmanship shall conform to IS:1081.The rate quoted shall also be inclusive of fixing windows, ventilators, louvers etc., in brick work, aluminum framing etc., by making holes, drilling holes in aluminum work wherever required, complete.

10.2.5 ALUMINIUM DOORS AND PARTITIONS ETC :

(a) SCOPE:

The contractor shall furnish all labour, material and equipment required for supply, transportation and fixing in position, all aluminum doors, windows and partitions including glass required for the civil works mentioned in this specification and as per drawings/directions.

(b) DOORS:

The door shall be built from anodized aluminum sections. The finished work shall be plumb, in line and free from corrugation, depression of wavy appearance and any defects. Field connections of all works may be made with concealed screws or approved type of fasteners or may be welded as approved by the engineer. All glazing shall be of high approved quality.

(c) PARTITIONS:

The partition shall be hollow aluminum section of approved size and gauge (with in-filled timber of silver oak) frame work, fully glazed or as per drawing. Glazing shall be fixed to the frame work with anodized aluminum sections and performed rubber beading all around.

11.0 TECHNICAL SPECIFICATION FOR SANITARY FITTINGS :

11.1 The Scope of work involves providing sanitary fittings of porcelain type in the toilets of the control room and providing ceramic tiles flooring in toilets.

11.2 Sanitary fittings to be provided are of standard make conforming to ISI specification and of approved colour. Following is the list of fittings to be provided in the toilets.

1. Water closet with flushing tank }
2. Wash Hand Basin } As per schedule
3. Flat back urinal }

11.3 All the above fittings are properly fixed in the flooring and walls with suitable means and neatly finished. Plumbing work for water line for toilets and bath

rooms and wash hand basin shall be done with 15mm dia. G.I. Pipe. Main line from Syntex tank shall be with 20mm dia G.I. Pipe line. W.C. line shall be laid with C.I. pipe line and connected to the main line in the man holes.

11.4 Bath room water and waste water from wash hand basin are taken separately and connected into a man hole located outside and later on connected to the pipe line to the septic tank. All the pipe joints are neatly finished with cement paste and buried in the flooring with necessary accessories like 'O' traps, gully traps and necessary CI/SWG bends. Care shall be taken that joints before covering with earth / concrete shall be tested for leakage and rectification carried out, if any, and closed.

11.5 The ceramic tiles of approved colour and make should be neatly fixed on the walls in toilets up to the level as shown in the approved drawing and also on flooring over a bed of cement mortar on the concrete. Care shall be taken while laying the ceramic tiles on flooring that proper slopes are maintained to avoid stagnation of water. The adequate downward slope shall be maintained while laying flooring tiles to drain out water to the gully traps. It is desirable to lay flooring tiles with mat finish to avoid slippery surface.

12.0 TECHNICAL SPECIFICATION OF YARD LIGHTING PILLAR BOX

Whether proof out-door type, 'Yard lighting Pillar Box for AC lighting having the following specifications for EHT sub-stations is to be supplied by the contractor

12.1 The box should be made of 2 mm thick Galvanised M.S. Sheet of size 700 x 500 x 150 (as per drawing enclosed). The box should have hinged type door vermin proof, weather proof, rubber beeding has to be provided around the door, door of the box should be provided with panel type lock. The top portion of the box shall be slanting, so that the rain water will fall away easily.

12.2 The box should be provided with 1 No. 60 Amps 3-Pole MCBs, 6 Nos. 20 Amps and 6 Nos. 10 Amps, Single pole MCBs and required copper Bus bars (20 mm x 5 mm) duly mounted Insulators fixed to Fiber glass sheet (neatly fixed in the box) with sufficient access for giving the cable connections. 20 Nos. cable glands should be supplied and the required holes are to be provided in the box duly half punched for taking cables into the box as per convenience. One hole should be of 35 mm dia for three phase cable and rest of 16 mm dia for lighting cables.

12.3 4 Nos. Galvanised M.S.Flat 50 x 6 mm should be welded to the box with one No. hole suitable for 5/8" dia bolts to facilitate easy mounting.

13.0 TECHNICAL SPECIFICATION FOR MARSHALLING BOXES FOR C.Ts, P.Ts, C.V.Ts.

The box should be made of 2 mm thick Galvanised M.S. Sheet of size 750x400x150 mm with hinged type door, vermin proof and weather proof. Rubber beading has to be provided around the door. Necessary brackets on the back side of the box should be provided for mounting on the structures. Doors of the box should be provided with panel type look. The top portion of the box shall be slanting so that rain water will fall away easily. Two rows having 20 Nos. terminals in one and 14 Nos. in other should be provided in the box. The terminals should be of disconnecting and testing type (Type CDTTS of 'connected make'. Three Nos. holes (Half punched) suitable for 6x2.5 sq.mm Copper cable and 3 Nos. holes (half punched) suitable for 4x2.5 Sq.mm copper

cable should be provided at the bottom to take the cables into the box. Suitable cable glands should be supplied. Terminals in the Marshalling box should have facility for easy earthing, shorting, isolating and testing. The following (as per drawing enclosed) are also required. 2 Sets of Earthing stud of 1/2" dia with suitable nuts, flat and spring washers are to be provided as shown in drawing.

For Marshalling boxes for PTs : Ten Nos. (10 Nos.) HRC fuses of 2 amps rating of M/s. English Electric Co., with neutral link (with Bolts and Nuts) and one No. 3 MCB (10 amps) (M/s. MDS make).

For Marshalling boxes for CVTs : 6 Nos. HRC Fuses of 2 Amps rating of M/s. English Electric Co., with neutral link with bolts and nuts.

14.0 TECHNICAL SPECIFICATION FOR MARSHALLING KIOSKS

The marshaling kiosk shall be of out door type suitable for operation in humid, dusty and tropical atmosphere with seasonal wet, dry and dusty condition as per clause-6.1 of I.S 5039/1969.The kiosk should be made of 3mm Galvanised M.S Sheet of size 1200x800x450mm.Each kiosk shall consist of (as per drawing enclosed.)

a) Detachable blank gland plate at the bottom as per drawing enclosed.

b) 200 Nos Heavy duty terminals stud type (5mm Sq) of rating 15 Amps to take 2.5 Sq mm copper and 4 Sq.mm Aluminum conductors.

c) 2 Amps HRC fuses (English Electric Company make)

d) Provision for three Numbers test terminal blocks.

e) Local illumination lamps - 2 Nos.

f) 2 /3- Pin 5/15 Amps. Multi point Plug with Switch - 3 Nos.

g) The terminal blocks shall be provided in vertical formation as shown in drawing. They shall be fully enclosed with removable covers made of barrier moulded integrally. Such block shall have washer and binding screw for external connection and white marking strip for ex-circuit identification and moulded plastic cover. All terminals shall be clearly marked with identification markers.

h) Arrangement of terminals shall be as to safely connect or disconnect terminals on live circuits and replace the fuse links.

i) Adequate space heaters shall be provided in the Marshalling kiosk as required.

j) The box shall be provided with self supporting Gavanised M.S. Channel 100x50mm.,welded to the bottom side of the Box (as detailed in drawing enclosed.) and 4 Nos. holes are to be provided (suitable for 3/4" dia bolts) for fixing to the plinth with galvanised bolts. The required 4Nos bolts of length 300mm, with galvanised nuts and washers are to be supplied.

k) The Kiosk shall be provide with a set of double hinged doors at the front and back. The doors shall be provided with 2Nos panel locks and 1No.pad lock as detailed in drawing. It shall have a door switch operated AC illuminated lamps which gets switched on when door is opened and gets switched off when closed. Two No. studs shall be provided on the body for giving earth connection to Marshalling Boxes.

15.0 TECHNICAL SPECIFICATION FOR ANTI -WEED TREATMENT AND ANTI - TERMITE TREATMENT:

15.1 ANTI-WEED TREATMENT: For safeguarding against the growth of weeds in the switchyard area weedicides (such as Biodex-C , Grammaxene, Fernoxene or equivalent products) should be mixed with water in accordance with the manufacturer's instruction (3 liters of weedicide mixed with 200 liters of water is a standard practice) and this solution should be spread over the earth with a chemical spraying pump on the ground.

The anti weed treatment should be given only after moisture content of the soil is brought below 29%. After completion of spraying of the anti-weed chemicals and before taking up the next activity, a period of 24 hours should be allowed for penetration of the chemical in to the soil.

15.2 ANTI-TERMITE TREATMENT :

15.2.1 SCOPE: The scope of work is to be set up a chemical barrier against attack by subterranean termites in the areas like cable trenches etc.

15.2.2 GENERAL: All work shall in general ,be executed as specified in IS:6313 part -II and as per approved specification of the agency having special know-how for the job. All necessary work to ensure uniform distribution and proper penetration of treating solution shall be done according to the instructions of the Engineer. Soil treatment shall not be done when it is raining or when the soil is wet with rain or sub soil water. Once formed, the treated soil barrier shall not be distributed .

15.2.3 CHEMICALS AND RATE OF APPLICATION : Any of the following chemicals (conforming to relevant Indian standards) in water emulsion shall be supplied by pressure pumps uniformly over the area treated.

Chemicals	Concentration / weight percent
Dieldrin (IS : 1052-1962)	0.5
Aldrin (IS : 1306-1958)	0.5
Chlordane (IS : 2863-1964)	1.0
Heptachlor	0.5

15.2.4 TREATMENT OF CABLE TRENCHES : Cable trenches may be fully enveloped by the chemical barrier by treating with chemicals at the rate of 15 liters /sqm on vertical surfaces of excavated trench walls and bottom surfaces at the rate of 5 liters / sqm. Chemical treatment shall be done in stages following the compaction of earth in layers. The treatment shall be carried out after the ramming operation is done by rodding the earth at 150mm centres close to the wall surface and spraying the chemicals in the specified dose.

15.2.5 TREATMENT OF EXPANSION JOINTS : These shall receive special attention and shall be treated in the manner approved by the Engineer.

15.3 ACCEPTANCE CRITERIA : The contractor shall give a one year service guarantee in writing supplemented by a separate and unilateral guarantee from the specialized agency for the period at no extra cost to the Employer.

15.4 I.S. CODE : Relevant code applicable for this specification : IS:6316 (part-II)-1971: Code of practice for Anti-termite measures in building.

16.0 TECHNICAL SPECIFICATION FOR PERMANENT STORM WATER DRAINAGE SYSTEM :

16.1 SCOPE :

The contractor shall furnish all labour, materials, tools and plants and equipment necessary for complete installation of permanent storm water drainage system and sewage disposal system required for the construction of the control houses 132kV / 220kV (new) and extension of 132kV/220kV both for main and ancillary works.

16.2 STORM DRAINAGE / RAIN WATER PIPES:

Rainwater PVC pipes shall be adequately provided to drain out the rain water without over flow. Minimum 110mm dia pipe(4Kg/Sqcm) shall be provided to drain out 40 sqm of roof surface area. All vertical down spout pipes and pipe fittings shall run inside the walls. All vertical down spout pipes shall be connected to water proof roofing by means of lead funnels ,protected at the out side by a cover net of brass wire or by 90 deg. bends(elbow).

PVC Pipes and Fittings: PVC Pipes where called for shall be either unplasticised PVC pipes or high density polyethelene pipes supplied with appropriate fittings. The plastic pipes are to be handled with care and stored in a place protected from sun. All plastic pipes and fittings shall be of approved make. Plastic pipes and fittings shall conform to ISI: 4984.

17.0 TECHNICAL SPECIFICATION FOR PILE FOUNDATIONS

17.1 BORED PILES : RCC CAST IN SITU (IS: 2911 Part-III): Installation Depth : The sizes and installation depths of the piles shall be as shown on the drawings. The piles, pile caps and plinth beams shall conform to the IS 2911 Part-III with RCC (1:2:4), (M20) grade concrete and necessary reinforcement. Laying of piles concreting shall conform to relevant clauses of APDSS.

17.2 TECHNICAL SUPERVISION :

The contractor's site staff shall include a person of adequate experience in the field of Geology who shall be in full time attendance during pilling operations. He shall be responsible for the positive and accurate identification of the rock horizon.

17.3 TOLERANCES :

17.3.1 **SETTING OUT :-** Setting out shall be carried out from the main grid lines of the proposed structures. Immediately before installation of the pile, the pile position shall be marked with suitable identification of pins or markers.

17.3.2 **POSITION :** For a pile head cut off, at or above ground level, the maximum permitted deviation of the pile head cut off center from the center point, shown on the setting out drawing, shall be 50 mm in any direction.

For a pile head cut off below ground level an additional tolerance will be permitted on the assumption that the pile head would have been within tolerance, if the cut off had been at ground level.

17.3.3 **VERTICALITY :** The maximum permitted deviation of the finished pile from the vertical shall be 1 in 100.

17.3.4 **FORCIBLE CORRECTIONS** : Forcible corrections to concrete piles shall not be permitted.

17.4 **PILING PROGRAMME :**

The contractor shall inform the Engineer each day of the programme of piling for the following day and shall give adequate notice of his intention to work outside normal hours and/or at week ends.

17.5 **DAMAGE TO PILES :**

The contractor shall ensure that damage does not occur to completed piles.

17.6 **RECORDS :**

The contractor shall keep records as indicated by table 1.1 given hereinafter of the installation of each piles and shall submit three signed copies of these records to the Engineer not later than noon of the next working day after the pile was installed. The signed records will form a record of the work.

Any unexpected driving or boring conditions, reported in accordance with particular obligations in this specification shall also be noted in these records.

TABLE 1.1. PILING RECORDS

a. Contract
b. Pile location and type reference
c. Specified cut off level
d. Ground level at commencement of pile installation
e. Working level (if difference from ground level)
f. As constructed diameter
g. Date and time of boring
h. Date and time of concreting
i. Levels at which seepages encountered
j. Bottom level and diameter of temporary casing
k. Level of top of rock together with details of sampling (level, length, type, description)
l. Pile toe level
m. Top of finished concrete level
n. Level of water or drilling fluid at commencement of placing concrete
o. Details of drilling fluid including specific gravity measurement as directed
p. Details of pile base cleaning operations
q. Full details of permanent casing
r. Length and details of reinforcement placed
s. Volume of concrete installed in pile
t. Concrete mix and control test results
u. Method of placing concrete
v. All information regarding obstructions, delays and other interruptions to the sequence of work.

17.7 **CONSTRUCTION OF PILES :**

17.7.1 **GENERAL** : Installation of piles shall be carried out in accordance with the pile construction drawings and as directed by the Engineer-in-charge.

17.7.2 **SEQUENCE OF INSTALLATION:** The sequence of installation of piles shall be subject to the approval of the Engineer-in-charge. In particular the sequence shall be selected to avoid piles being bored close to other recently constructed piles.

17.7.3 **MATERIALS:** Drilling fluid (Bentonite) shall be in accordance with clause A 3.1 of IS 2911 Part-I, Section-II.

17.7.4 **EQUIPMENT AND ACCESSORIES:** The equipment and accessories required for construction of bored piles shall be of a standard type and

approved by the Engineer.

17.7.5 **BORING:** While boring in cohesionless deposits below the water table care shall be taken to avoid the inflow of water and soil into the bore hole by adding water and maintaining a level potentially above the standing water level. If the hole is unstable during boring either temporary casing or substitution of drilling fluid for the water shall be used to ensure stability. The method of stabilising the hole shall be approved by the Engineer. In cohesive soils, water shall not be added for boring purposes.

17.7.6 **STABILITY OF PILE EXCAVATION USING TEMPORARY CASINGS:** Temporary casing shall be advanced with the boring tools, and excavation shall not be carried out below the toe of the casing.

Temporary casings shall be free from significant distortion. They shall be of uniform cross section throughout each continuous length. They shall be free internal projections and encrusted concrete which might prevent the formation of piles.

17.7.7 **STABILITY OF PILE EXCAVATION USING BENTONITE SLURRY OR EQUIVALENT:** The level of the fluid in the excavation shall be maintained so that the fluid pressure always exceeds the pressures exerted by the soils and external ground water, and an adequate temporary casing shall be used in conjunction with the method to ensure stability of the strata near ground level in the base shall be maintained at a level not less than 1 mtr. above the equilibrium level of the ground water. During boring the consistency of the drilling fluid shall be controlled in accordance with IS-2911 part-I, section-II.

17.7.8 **SPILLAGE AND DISPOSAL OF DRILLING FLUID**: All reasonable steps shall be taken to prevent the spillage of bentonite suspension on the site in areas outside the immediate vicinity of boring. Discarded bentonite shall be removed from the site without delay. Any disposal of bentonite shall comply with the regulation of the local controlling authority.

17.7.9.1 **PUMPING FROM BORE HOLES**: Pumping from bore hole shall not generally be permitted. However small quantities of accumulated water may be pumped from dry bore for cleaning out purposes prior to concreting.

17.7.10 **CONTINUITY OF CONSTRUCTION**: Boring of any pile shall not be commenced unless it can be completed and concreted within the same working day.

17.7.11 **CLEANLINESS OF PILE BASES** : On completion of boring, loose, disturbed or remoulded soil and rock shall be removed from the base of the pile.

17.7.12 **REINFORCEMENT OF PILE** :The main longitudinal reinforcing bars in piles shall be in one continuous length unless otherwise approved. In long piles, joints will be permitted in main longitudinal bars but these shall

be kept to a minimum. Joints in adjacent bars shall be staggered at least 1m apart along the length of the pile. Lapped joints shall be staggered by atleast 1.3 times the laps length. Joints in reinforcement shall be such that the full strength of the bar is effective across the joint and the reinforcement cage is of sound construction for installation in the pile bore.

Longitudinal reinforcement shall be provided for the full length of the pile and shall extend above pile cut-off level as shown on drawings. Longitudinal reinforcement shall be formed into a rigid cage to resist deformation during handling and installation if necessary by the use of additional reinforcement subject to the approval of the Engineer.

Lap or splice joints shall be provided with sufficient link bars to resist eccentric forces.

17.7.13 **COVER** : Minimum cover to main reinforcement in the pile shaft shall be 75 mm and should be subject to approval of Engineer.

17.7.14 **CONCRETING GENERAL** :The method of placing and the workability of the concrete shall be such that a continuous monolithic concrete shaft of the full cross section is formed. Mechanical vibrators shall not be used to compact concrete.

17.7.15 **CONCRETE MIX PARAMETERS** : Concrete in piles shall be of grade M15 except that its consistency shall be compatible with the particular method of pile installation proposed. The slump of the concrete shall range between 100 and 180 mm depending on the manner of concreting in accordance with clause 6.34 of IS 2911 Part I.

17.7.16 **CONCRETING PLACING** : The placing of concrete which shall not commence until the Engineer has approved the bore bole, shall be completed as soon as possible.

The concrete shall be placed in such a manner that segregation does not occur. Concrete to be placed below water or drilling fluid shall be placed by tremie or other approved method and shall not be discharged freely into water for drilling mud. Where tremie placing is to be used it shall be in accordance with clause 7.5 of IS 2911 Part I. Section 2. Where concrete is placed through drilling mud, the specific gravity of the mud at the bottom of the boring shall not exceed 1.2 immediate prior to placing concrete. The amount of mud to be sampled should be adequate to suit he test apparatus (generally ½ to 1 litre is adequate.)

17.7.17 The cut off level shall be as shown on the construction drawings. Concrete shall be cast to a minimum of 500 mm above the cut off level or local ground water level whichever is the higher.

17.7.18 **EXTRACTION OF CASING** : The method of extracting the temporary casing shall be to the approval of the Engineer in charge and shall be such that the concrete in the casing does not lift and maintains its workability so tht it flows freely against the pile bore. Utmost care shall be exercised in maintaining an adequate head of concrete above the bottom

of the casing until concreting is completed so as to prevent the inflow of soil and/or water into the bore.

17.7.19 **TEMPORARY BACKFILLING** : Sections of empty hole remaining above the head of the pile shall be temporarily backfilled with sand of a distinctive colour prior to excavation for pile cap construction.

17.7.20 **REPLACEMENT OF DEFECTIVE PILES** : The Engineer reserves the right to reject any pile which is defective in any way. Defective piles shall be treated as directed by the Engineer . One or more additional piles to substitute the defective pile shall be installed at the contractor's cost as directed by the Engineer.

17.8 LOAD TESTS ON PILES :

17.8.1 GENERAL: The load tests shall be carried out by the contractor as described herein which shall be paid for saparately. These tests however, shall not normally be undertaken by the contractor within 28 days after the installation of the pile in position. The load tests shall be carried out on single pile. Before any load test is made the proposed arrangement for carrying out the load tests including the preparation of the structure to receive the load and the type of loading to be adopted shall be furnished by the contractor and approved by the Engineer .The load tests shall be made under the supervision of the Engineer. The complete record of such loads shall be filed with the Engineer, in triplicate . The responsibility for carrying out such load tests satisfactorily and safely and on proper lines rests with the contractor.

17.8.2 **METHOD OF TESTING :**

The test shall be conducted as per ISS 2911 (Part I – Para D4 Cyclic – loading tests) except as described herein.
The test load to be applied on the loading platform supported on pile shall be twice the proposed safe designed load of the pile. The test load shall be applied in 7 increments equal to ¼, ½, ¾ 1, 1 ¼ , 1 ¾, and 2 times the proposed working load. Readings of the settlement and rebounds shall be reckoned against a constant elevation bench mark and shall be recorded to $1/30^{th}$ of a cm. for each increment or decrement of load. After the proposed work load has been applied and for each increment thereafter the test load shall remain in place until there is no settlement in a two hour period. The final test load shall remain in place until the settlement does not exceed $1/30^{th}$ of a cm. in 48 hours. The final test loading shall be removed in decrements not exceeding $1/4^{th}$ of the total test load with intervals of not less than 1 hour rebound shall be recorded after each decrement is removed and the final rebound shall be recorded 24 hours after the entire test load has been removed. The complete record in triplicate shall be filed with the Engineer on the loads and readings obtained duly verified and countersigned by the Engineer's representative.

17.8.3.1 **NUMBER OF TESTS:** Not less than 2 (two) piles should be subjected to load tests for 100 Nos. of piles installed. The piles to be tested shall be selected from separate groups or as indicated by the Engineer. Further

tests, if necessary shall be made on the basis of the results of load tests. The test piles shall be used for foundation of the structure.

17.8.3.2 **NET PERMISSIBLE SETTLEMENT** : Net settlement of the top of pile after the conduct of cyclic tests with the maximum load of twice the design load shall not be greater than 12 mm.

17.8.3 **PULL OUT TEST**: A trial pile shall be installed in such a location as directed by the Engineer , such that it does not interfere with working piles. The specification and details of the trial pile shall be same as for working pile. An outer casing shall be provided such that the trial pile has a clearance of atleast 20mm alround. The object of the test is to determine the strength of socketting provided for in the drawing and adjustments, if any required. The test shall be carried out to failure point.

17.9 **METHOD OF MEASUREMENT** :

Forming pile shaft shall be measured in running metres from the approved toe level to cut-off level for each diameter of pile.

Empty boring shall be measured in running metres from working ground level to cut-off level for each diameter of pile.

Concrete pile shaft shall be measured in running metres from the approved toe level to cut-off level for each diameter of pile.

Reinforcement including projection above cut-off level is measured by weight in tonnes. The weight shall be arrived at my multiplying the length, shown on the drawings or otherwise directed by the Engineer – in charge by the sectional weights.

Trial piles, reactions piles, rock chiselling if any shall be measured and 'clubbed together' with works piles, although tests are enumerated separately.

IS CODES (LATEST EDITIONS)
Important relevant IS in this section.

IS 2911 – Part I Section	:	Concrete piles section bored cast in situ piles.
IS 2911 – Part IV	:	Load test on piles.
IS 2131	:	Method for standard penetration test for soils.

17.10 UNDER REAMED PILES.

The execution of under reamed pile foundation shall conform to IS code 2911 (Part III) 1978 and specification given in building Digest No. 56 of Central Building Research Institute Roorkee. The option is open to Transmission Corporation of Andhra Pradesh Limited and contractor is bound to execute any or all capacity of piles.

The depths of piles are tentatively assessed with double bulb. The depths are likely to be increased or decreased based on site conditions. No extra

rate can be entertained in this account and only proportionate relevant unit rate will hold good as deduceable from the formule given below for double bulb

$$= \frac{\text{Agreement rate } (L+12d)}{X + 12\ d.}$$

Where L = Actual length of piles.
D = diameter of pile
X = standard length as per Schedule A

17.11 PILE CAPS AND FOUNDATIONS

17.11.1 DETAILED PROGRAMME

The contractor shall prepare a detailed programme showing his capacity for pouring concrete and locations of the positions of construction joints , if any in the beams, slabs etc. These drawings should be approved by the Engineer in charge and the construction joints shall be maintained exactly at location as shown in these approved drawings.

17.11.2 GENERAL

The necessary concrete work in pile caps, footings, tie beams, pedestals, wall slabs etc., including reinforcement and from work have to be done by the contractor as shown in the construction drawing and as per specifications for concrete.